RPMGLOBAL

AMT - Work Execution and Finalisation

Training Manual

Copyright © 2024 RPM Software Pty Ltd. All Rights Reserved



RPM SOFTWARE – ASSET MANAGEMENT

Table of Contents

Introduction to AMT **6**
Process **7**
Training Workshop (Perform Planned Work) **13**
1. Receive Documentation 13
2. Assign Work 14
3. Collect Job Instructions, Parts and Tooling 14
4. Update AMT Live 14
5. Perform Planned Work 16
6. New Defects Identified – Perform Now? 16
7. Job Preparation 16
8. Perform Additional Work 19
9. Record Defect 19
10. Release the Asset to Operations 21
11. Complete an Event in Shiftlog 21
12. Completed Job Folders 22
13. Validate Documentation 22
14. Daily Downtime and Work Order Capture 22

Training Workshop (Perform Breakdown Work) **23**
15. Receive Notification 23
16. Dispatch Mechanics 23
17. Log the Asset Down 23
18. Diagnose Problem 26
19. Authorisation 26
20. Perform Planned Work 26
21. New Defects Identified – Perform Now? 26
22. Job Preparation 27
23. Perform Additional Work 27
24. Record Defect 27
25. Release the Asset to Operations 27
26. Set AMT Breakdown Event to Completed 29
27. Complete Job Folder 30
28. Validate Documentation 30

Training Workshop (External Downtime) **31**
29. System Configurations and Settings 31
30. Downtime Interface Process 32
31. Managing Downtime Notifications 34

Training Workshop (Work Finalisation) **36**
32. Perform Planned and Breakdown Work 36

33. Enter New Defects into AMT 36
34. Update AMT with Downtime and Feedback Information 36
35. Provide Job Folders in Asset Files 42
36. Record Additional Work Order Details 42
37. File Job Folders in Asset Files 52
38. Reconcile Downtime to Dispatch 52
39. Enter Employee Timesheets 52
40. Produce a Daily Log Report for the Daily Production Meeting 54
41. Review and Confirm Events 56
42. Work Order Settlement Closure 58

Training Workshop (Usages, Measurements and Production Statistics) 62

43. Actual Usages 62
44. Measurements 65
45. Production Statistics 68

Training Workshop (Product Incident Capture & Warranty Claims) 73

46. Registering Product Incident Reports 73
47. Reviewing Product Incident Reports 74
48. Assess Warranty Claims 75
49. Submit Warranty Claims 78
50. Not Claimable Warranty Claims 79

Preface

The training workshops are developed to supplement instructor-led training. This guide is constructed in a way that teaches students the key functionality of the application through practical, hands-on workshops. Each workshop details tasks that the student is required to complete. The workshops build on each other to help ensure a comprehensive understanding of the application is achieved at the completion of the course. Space is provided at the end of each workshop for learning checks and student notes.

Symbol Library



Note

A note for participants about the task.



Important Reminder

An important reminder for the participant about the task.



Learning Check

Check your understanding of the workshop and ask any related questions.

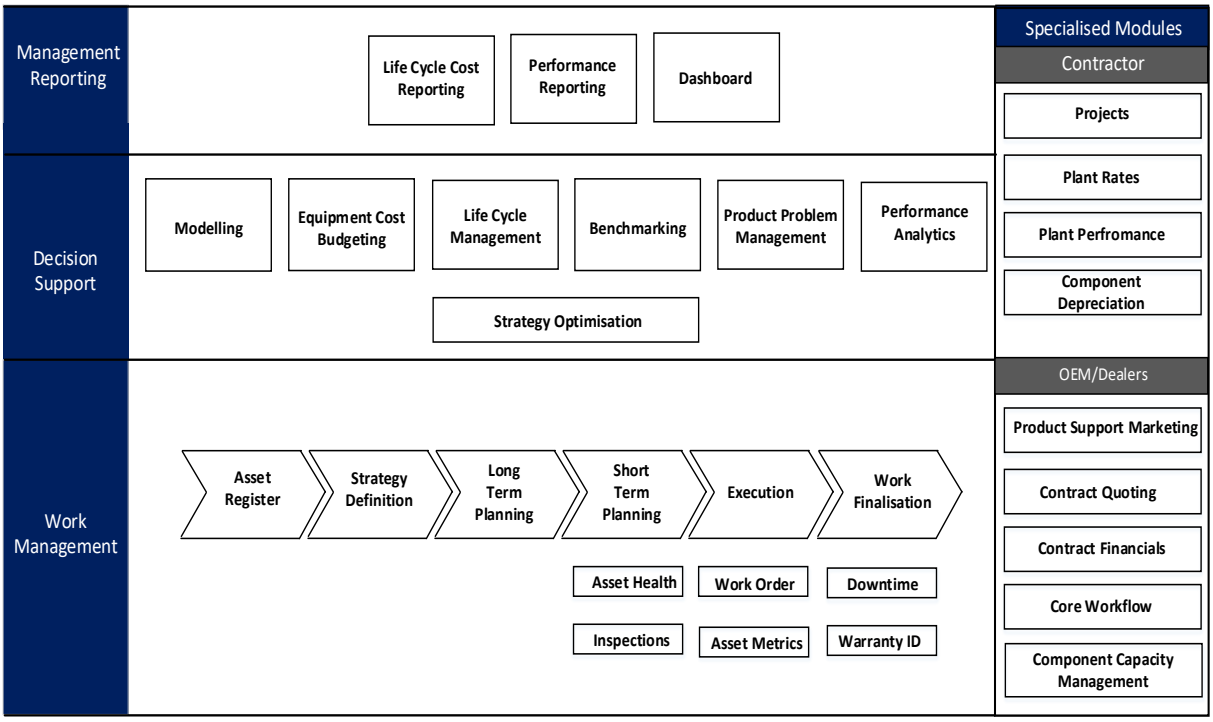
Course Outline

Prerequisites	Getting Around AMT, AMT Key Concepts.
Target Audience	Store Coordinator, Reliability Engineer, Workshop Mechanic, and Site Clerk
Nominal Duration	
Course Objectives	The objectives of work execution are: • Perform planned work efficiently, safely, and to a high degree of quality. • Record maintenance history and labour allocations accurately. • Record new defects identified. The objectives of work finalisation are: • Capture information about the downtime and work order accuracy • Review and close work orders

Learning Outcomes	At the end of the course, participants will have learned to: • Perform a breakdown from the shift log o Add a new in-progress breakdown downtime event o Set all the event and work order fields o Add an additional work order to the event directly in the shift log o Complete the event • Perform planned work from the shift log o Select a YTS event o Set the YTS event into In Progress o Complete work orders o Complete the event • Allocate downtime o Open a completed event from the shift log o Add downtime allocations o Assign some of the downtime to the customer (or operations) for cleaning the machine • Add a new event from the Events tab o Create a breakdown event o Complete the breakdown event o Allocate the downtime • Review events from the Equipment Down tab o Review planned events in the next month • Complete work order technical information o Open a completed work order and complete the feedback form o Set the component serial number for a rotable item o Review the changeout indicators (EOL, Partial, Swing) o Specify symptom, cause, repair information and occurrence type o Record warranty claims • Update labour hours in timesheets o Record actual labour hours using the timesheets • Update usages, measurements and statistics o Record actual usage readings o Record measurement readings o Record production statistics • Run and understand daily reports o Update product incident reports
Assessment Strategy	The completion of workshop tasks.
AMT version	8.26

Introduction to AMT

AMT is a comprehensive asset management system consisting of integrated modules as depicted below. The functionality covered by this training manual is identified by the Execution and Work Finalisation sections in the diagram.



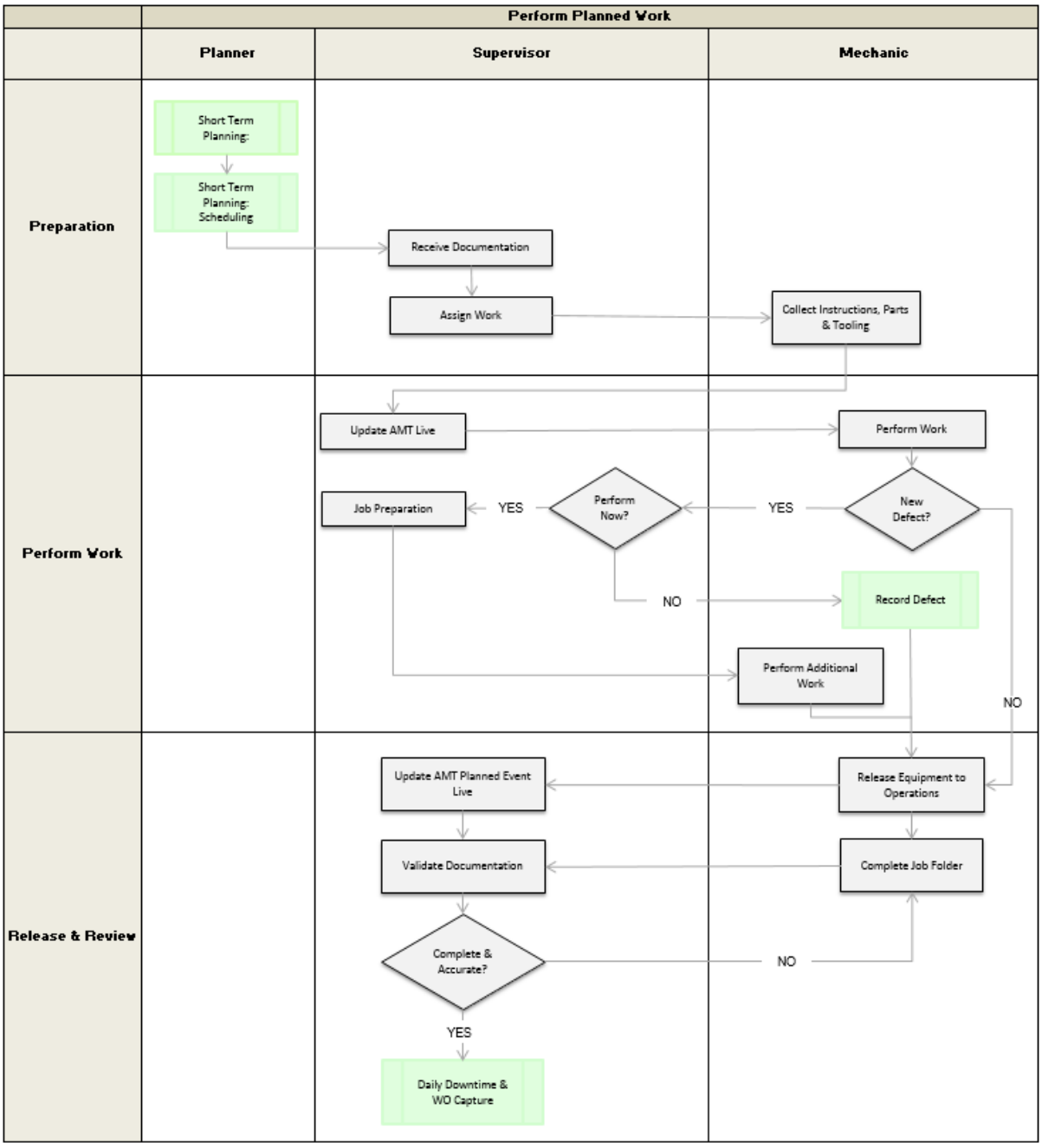
Work execution is the process of performing work. Work execution is broken into three distinct sub-processes:

- Perform planned work - the process of executing work that has been planned.
- Perform breakdown work - the process of executing breakdown work.
- Work finalisation - the process of capturing repair information and closing work orders.

Finalisation is the process of ensuring actual work order data has been captured correctly.

Process

PROCESS DIAGRAM (PERFORM PLANNED WORK)



PROCESS INPUTS (PERFORM PLANNED WORK)

The following inputs are required for this process:

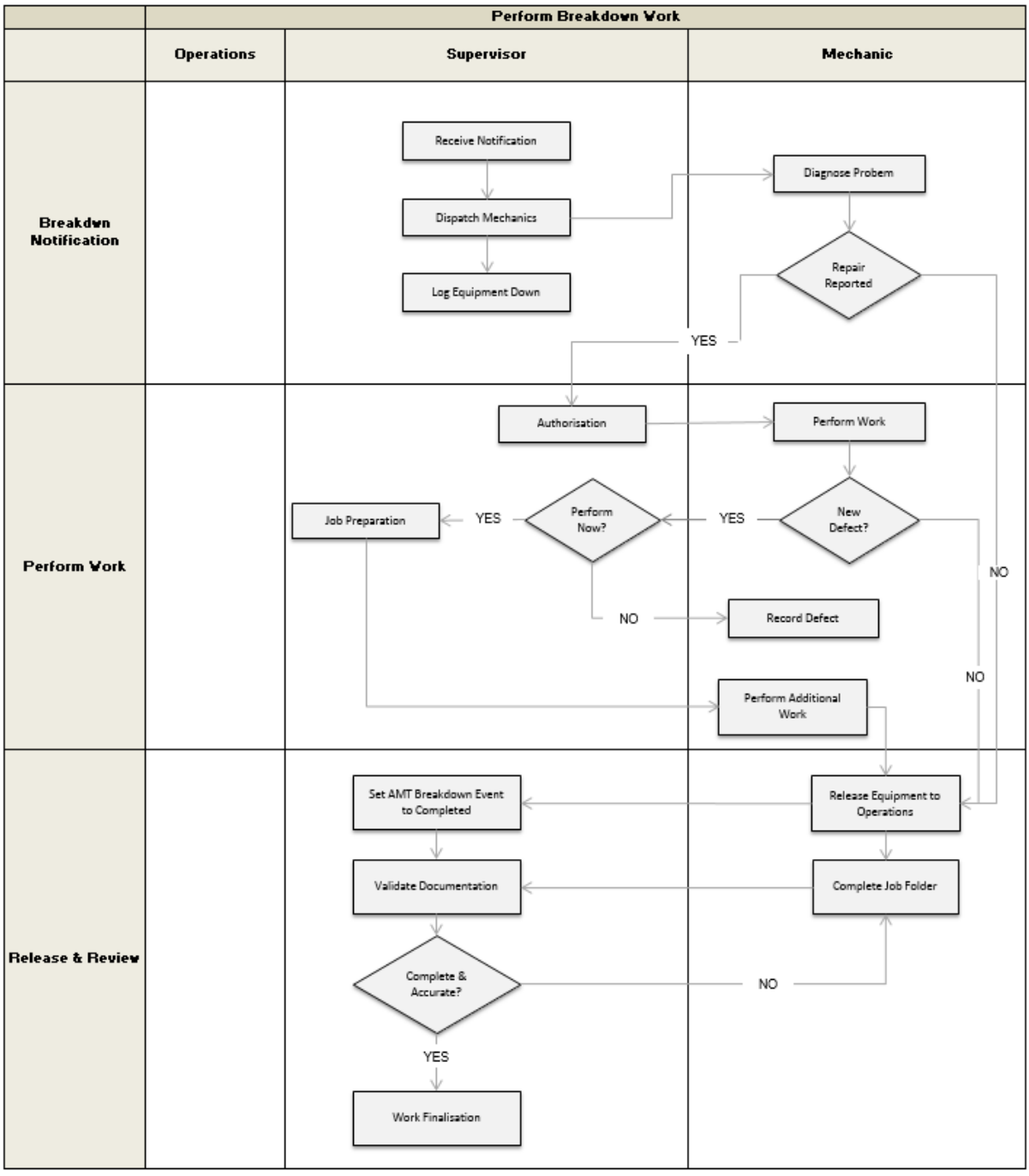
- Job folders (instructions).
- Resources available.

PROCESS OUTPUTS (PERFORM PLANNED WORK)

The following outputs can be expected from this process:

- Completed job folders.
- Defects captured.

PROCESS DIAGRAM (PERFORM BREAKDOWN WORK)



PROCESS INPUTS (PERFORM BREAKDOWN WORK)

The following inputs are required for this process:

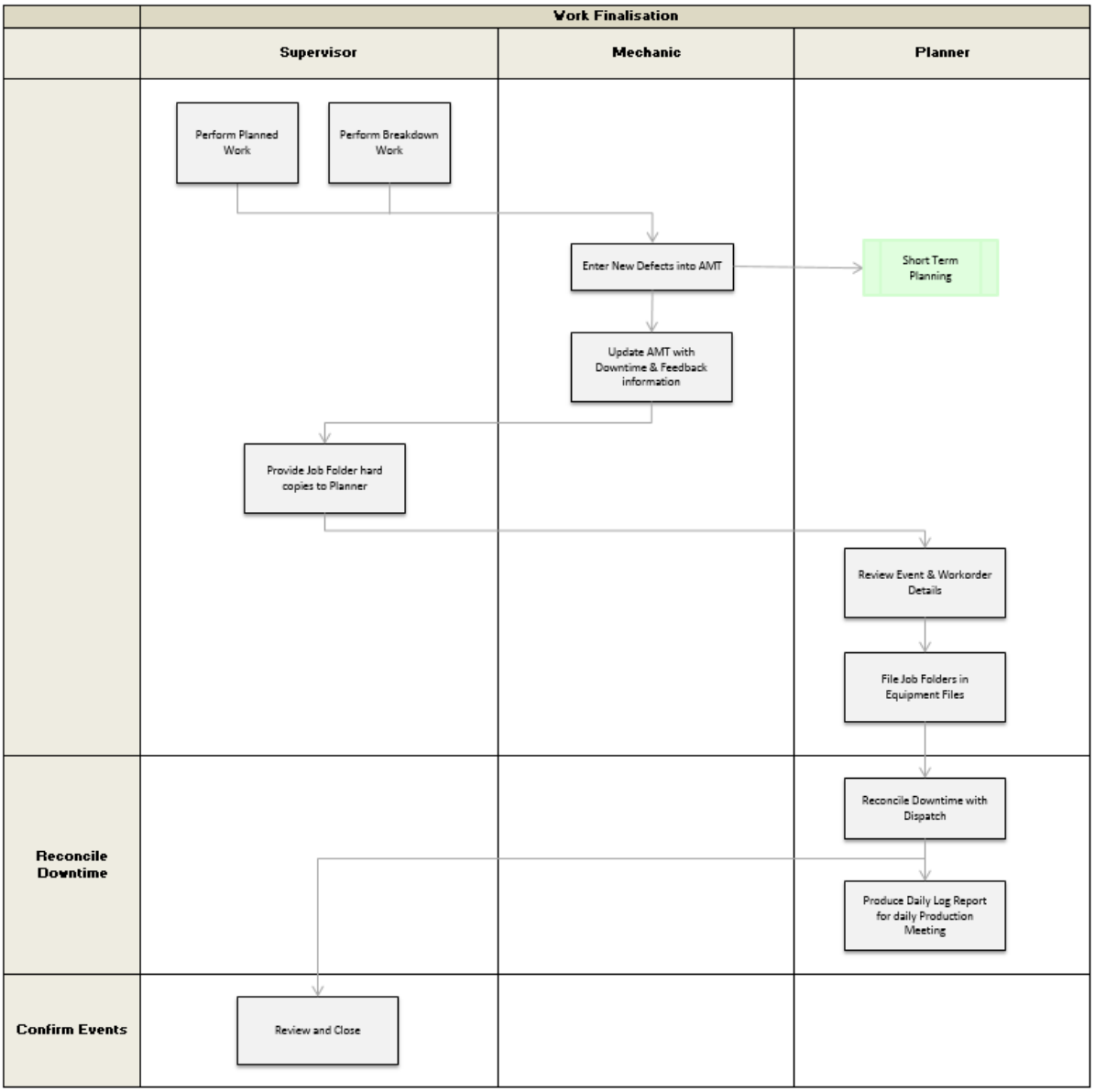
- Resources available.

PROCESS OUTPUTS (PERFORM BREAKDOWN WORK)

The following outputs can be expected from this process:

- Completed job folders.
- Defects captured.

PROCESS DIAGRAM (WORK FINALISATION)



PROCESS INPUTS (WORK FINALISATION)

The following inputs are required for this process:

- Event and work orders.
- Completed job folders.

PROCESS OUTPUTS (WORK FINALISATION)

The following outputs can be expected from this process:

- Maintenance history.
- Availability and reliability KPIs.

Training Workshop (Perform Planned Work)

1. Receive Documentation

Planned work is organised by planners following the **Short-Term Planning** process. The outputs of short-term planning are:

- Work plans.
- A job folder consisting of:
  - **Eventscope** – details all tasks performed during the event (asset stoppage).
  - **Workscope** – task operations with parts and labour allocations.
  - **Detailed task instructions** or references to product handbooks.
  - **Checksheets** – safety, environmental and statutory.
  - Permits and certificates.
  - **Timesheets** – to record the time spent on the job (this will vary based on the different payroll systems).
  - **Data capture forms** for condition monitoring.
  - Bill of materials (BOMs).
  - Defect listing.
- Parts ready for collection.
- Specialised tooling ready for collection and other workshop facilities arranged.

2. Assign Work

From the work plans, the supervisor assigns the work to the mechanics, ensuring they understand the scope and are competent to perform the work.

The supervisor needs to ensure the work plans and labour allocations are clearly visible; this could be on a whiteboard or in AMT.

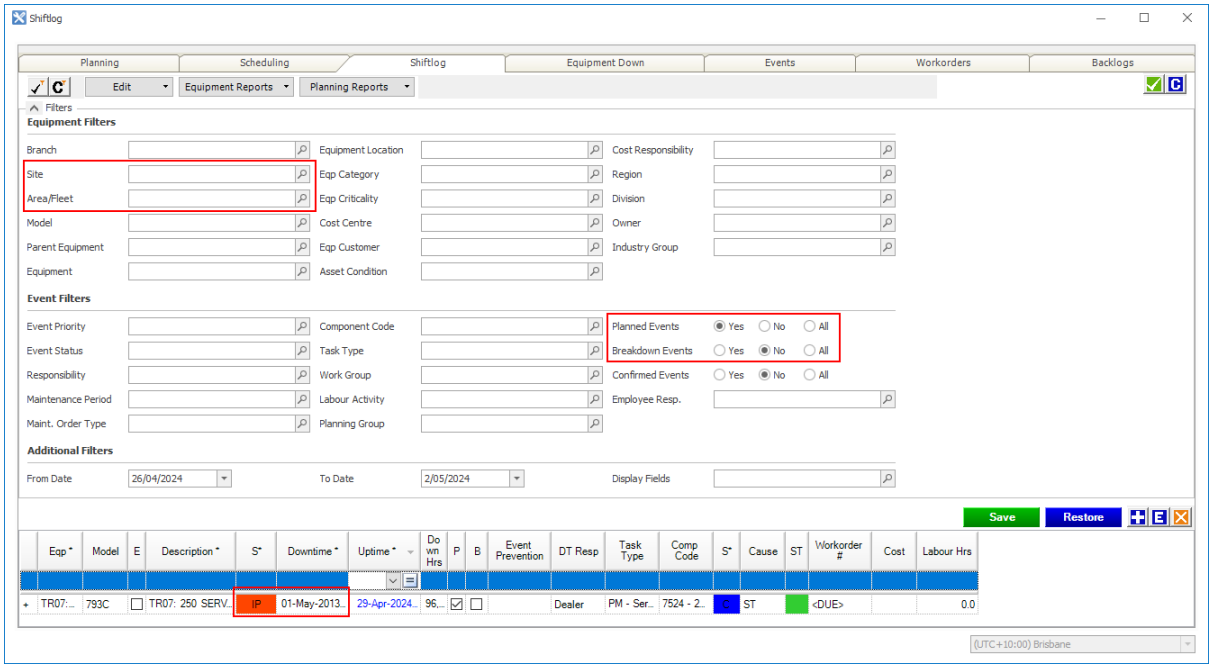
3. Collect Job Instructions, Parts and Tooling

Mechanics collect the job folders, parts and tooling from their designated areas.

4. Update AMT Live

Procedure – Update AMT Live

1. If AMT is being updated live, the supervisor updates the existing planned event in AMT using the **Daily Operations** > **Equipment Status > Shiftlog** tab.
2. Filter for the **Site** and **Area/Fleet**.
3. Set **Planned Events** to **Yes**, and **Breakdown Events** to **No**.
4. Click **Apply Filters**.
5. When the work begins, set the **Status** to **In Progress**.
6. Set **Downtime** to the time the asset actually went down.
7. Click **Save**.



Exercise – Set a Planned Event to "In Progress"

1. Find a planned (YTS) event in Shiftlog.
2. Change the event status to In Progress.
3. Record the actual downtime and set the primary work order status to In Progress.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

5. Perform Planned Work

Functionality Overview

Mechanics perform the work, ensuring all safety and environmental procedures are followed.

6. New Defects Identified – Perform Now?

If a new defect is identified, the mechanic should diagnose the problem and consult the supervisor whether it should be repaired immediately or registered to be performed later. Factors in making this decision include:

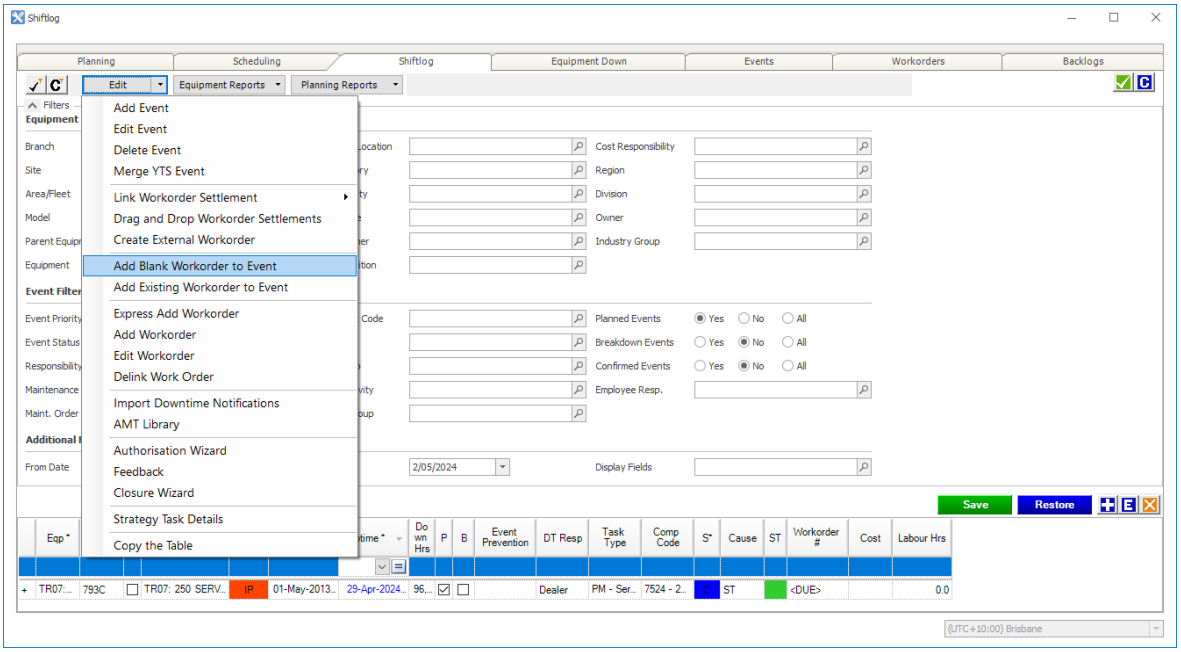
- If deferring the repair could significantly affect safety, environmental, cost or production matters, the repair should be performed immediately.
- Is it efficient to perform the repair later? Consider the impact on production, whether authorisation is required, whether resources (parts, labour, tooling) are available and when would be the most efficient time to do the work.

7. Job Preparation

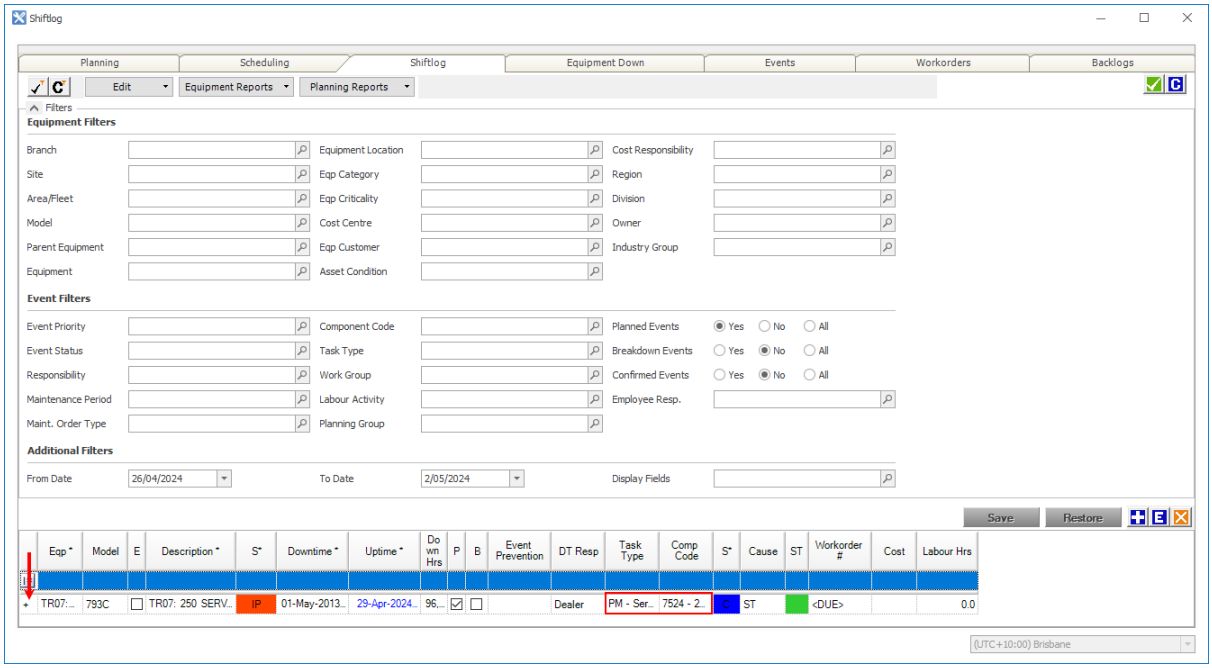
Procedure – Job Preparation

Use **Shiftlog** to add a new work order to the existing planned event.

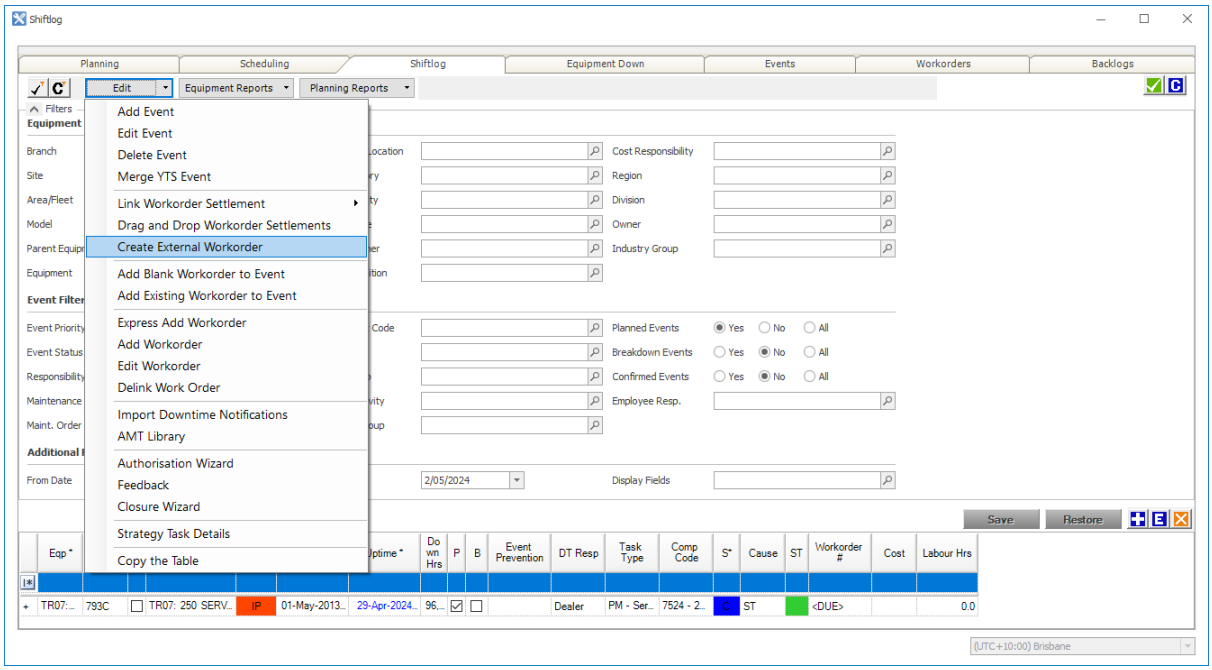
1. Click **Edit** and select **Add Blank Workorder** to Event



4. Enter work order details in the **Shiftlog**.
   a. Click **+** to expand the event.
   b. Enter the **Task Type** and **Component Code**.
   c. Click **Save**.



5. Create a new external work order (which is a work order in your ERP system, by clicking **Edit** > **Create External Workorder**. Assign parts and labour to it and obtain tooling. This will not generate an ERP work order if the interface is not in place.



If parts are not required, the work order could be opened in the external system later.

Exercise – Add a New Work Order to an Existing Event

1. Find the event used in the previous exercise in the Shiftlog.
2. Add a second work order to the event (for example, a work order to correct a cracked mirror).
3. If there is an external work order system, create the external work order and verify that it is created correctly (for example, correct asset, correct coding).

8. Perform Additional Work

Mechanics execute any additional defect work and record the results in the documentation.

9. Record Defect

If the defect is to be recorded and performed later, either:

- record the defect details on the defect capture form.
- enter the defect directly into AMT.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

10. Release the Asset to Operations

When work is almost complete, the mechanic should inform operations so that an operator can be ready to start the asset when it's time.

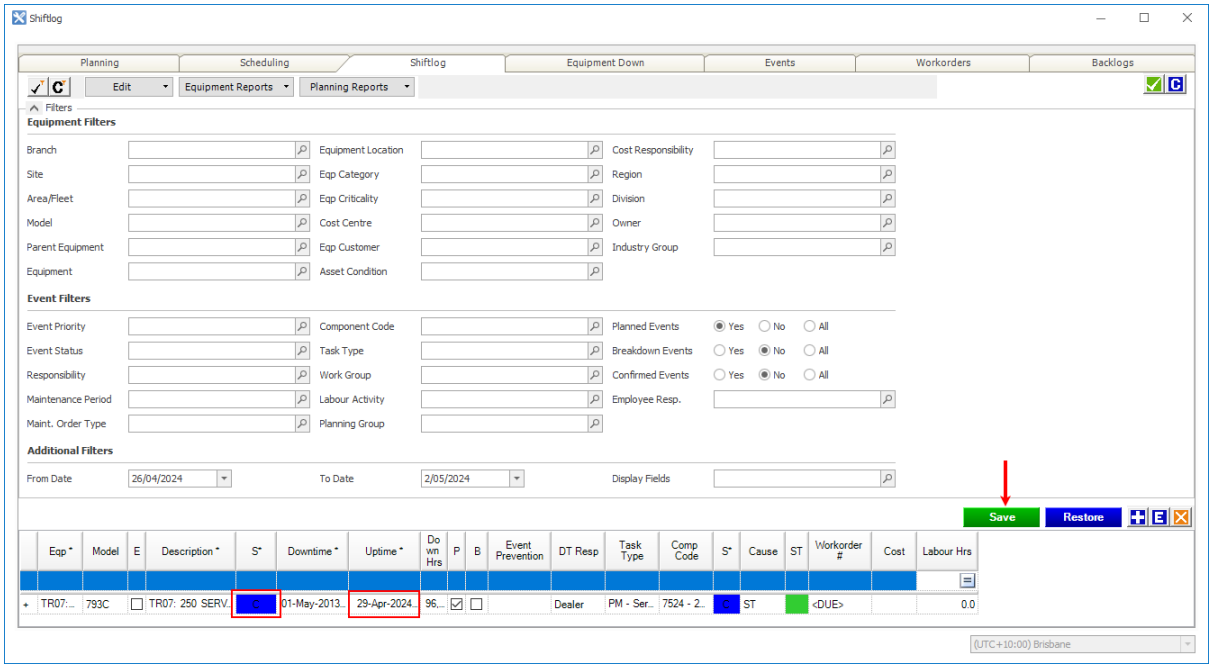
When the work is complete and the asset is ready to resume work, the mechanic informs operations.

11. Complete an Event in Shiftlog

Procedure – Complete an Event in Shiftlog

If AMT is being updated live, the supervisor updates the existing planned event using **Equipment Status** > **Shiftlog**.

1. Set the event **Status** to **Completed**.
2. Enter the actual **Uptime**.
3. Click **Save**.



Exercise – Completing an Event in Shiftlog

1. Find the event used in previous exercises.
2. Complete the work orders and the event, and record the actual uptime.

12. Completed Job Folders

Functionality Overview

The mechanic should complete the following documents:

- **Eventscope** – record the downtime details and update the work order status (if completed) and the duration of each work order.
- **Work order** – record the repair notes and sign-off details on the workscopes.
- **Labour** – depending on the processes followed, the mechanics may complete timesheets or enter labour details directly onto the workscopes.
- **Parts Return Advice** – details of parts to be returned to stores.

When you return to the workshop, give all documentation (including defect capture forms from above) to the supervisor.

13. Validate Documentation

The supervisor reviews the documentation supplied by the mechanics and checks that it is complete and accurate. If it is not complete or is inaccurate, the supervisor gives the documentation back to the mechanic to complete correctly.

14. Daily Downtime and Work Order Capture

Complete fields in the event and work orders.

Training Workshop (Perform Breakdown Work)

15. Receive Notification

Depending on the nature of the operation, the notification that the asset is down may come from different sources (operator, dispatch, field mechanics, and so on). It is important that it is clearly defined who should receive the notifications and the reporting channels. Generally, the field supervisor/foreman receives the notification.

16. Dispatch Mechanics

This will depend on the nature of the breakdown, priority, skills required, availability of resources, and so on.

17. Log the Asset Down

Procedure – Log the Asset Down

The downtime and initial breakdown details should be recorded in AMT as soon as possible. One of these two approaches is generally followed:

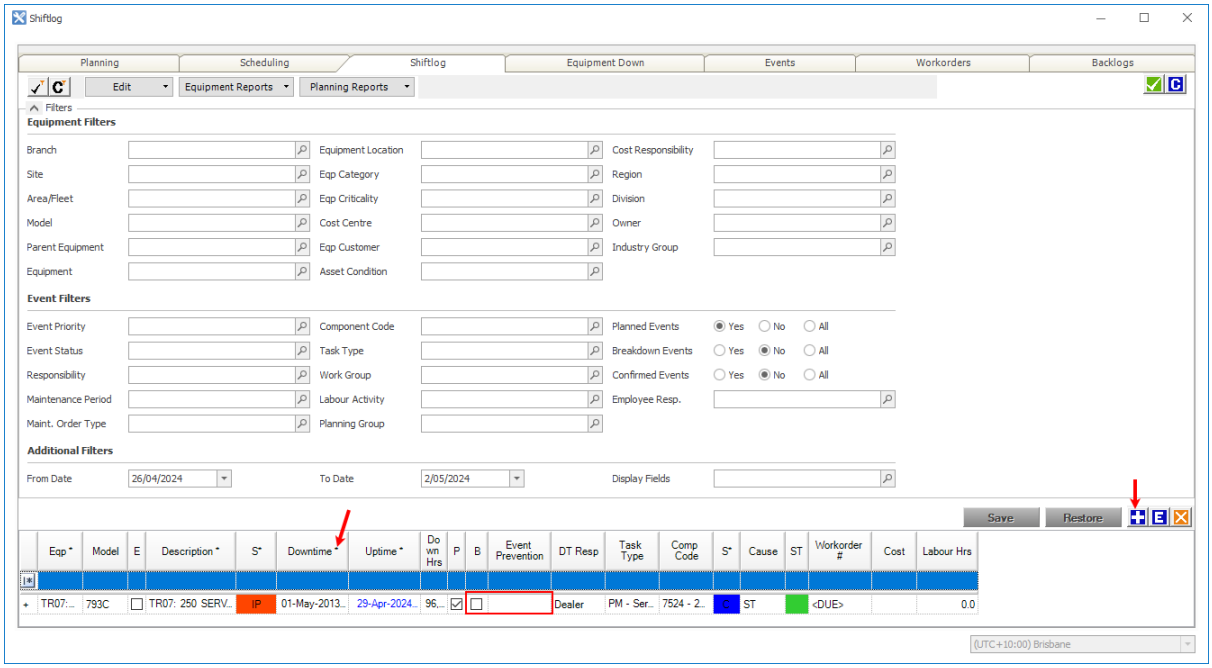
1. Record the details on a paper-based shiftlog and update AMT the following day.
2. Record the details directly into AMT (**PREFERRED**).

If AMT receives a downtime notification from an external dispatch system (such as MineStar, Jigsaw, or Modular Mining), the notifications appear in the Shiftlog as events (unless AMT has been configured to place these notifications in a temporary holding area called **External Downtime Wizard**).

If the external event represents a planned event, it should be merged with the existing planned event (already created by the planner).

If AMT does not receive downtime notifications for the asset from an external system, the supervisor will create a new In Progress event in AMT using **Shiftlog**.

1. Click **Add**.
2. Complete the mandatory fields (***** in the column header).
3. Select the **B** (Breakdown) check box and filter for an **Event Prevention** reason.



Exercise – Log an Asset Down

Log that an asset is currently down for a breakdown event using Shiftlog.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

18. Diagnose Problem

Mechanics diagnose the cause of the breakdown and the scope of work to perform the repair.

19. Authorisation

Every organisation has different authorisation processes. For breakdowns, this may be verbal, signing a form, or authorising the work order. AMT work order authorisation is covered in detail in the **RPM Training – AMT Short-Term Planning** manual.

20. Perform Planned Work

Mechanics perform the work, ensuring all safety and environmental procedures are followed.

21. New Defects Identified – Perform Now?

Functionality Overview

If a new defect is identified, the mechanic should diagnose the problem and consult the supervisor whether it should be repaired immediately or registered to be performed later. Factors that influence this decision include:

- If deferring the repair could lead to a significant impact on safety, environmental, cost or production matters, the repair should be performed immediately.
- Is it efficient to perform the repair later? Consider the impact on production, whether authorisation is required, whether resources (parts, labour, tooling) are available and when would be the most efficient time to do the work.

,

22. Job Preparation

Functionality Overview

If the defect is to be performed now, the standard planning process needs to be followed:

- **Work order instructions**: Are the appropriate work order instructions and checklists available? These could either be generic documents attached to the instructions or instructions detailed on the defect capture forms.
- **Parts available:** Are all the required parts in stock? Parts not in stock will need to be ordered and their lead times identified.
- **Specialised labour**: Is specialised labour required that may not be readily available? If so, the lead time and availability dates will need to be ascertained.
- **Materials**: Are there any additional materials or consumables that need to be ordered to complete the work?
- **Outwork**: Is any outwork (sub-contract) required that may not be readily available? If so the lead time and availability dates will need to be ascertained.
- **Tooling**: Is any specialised tooling, facilities, vehicles or craneage required that may not be readily available? If so the lead time and availability dates will need to be ascertained.
- **Work Permits**: Are required permits for hot work, confined space work, and working at heights completed and authorised for the work orders?
- **Workorder**: Has an ERP work order been opened for the work order? (**Note**: This only applies if AMT is interacting with another work order system).



*If the work order is to be created directly in AMT, refer to the **RPM Training – AMT Short-term Planning** manual.*

23. Perform Additional Work

Mechanics execute the additional defect work and record the results in the documentation.

24. Record Defect

If the defect is to be deferred, record the defect details on the defect capture form.

25. Release the Asset to Operations

When work is almost complete, the mechanic should inform operations so that an operator can be ready to start the asset when all work is completed.

When the work is complete and the asset is ready to resume work, the mechanic informs operations.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

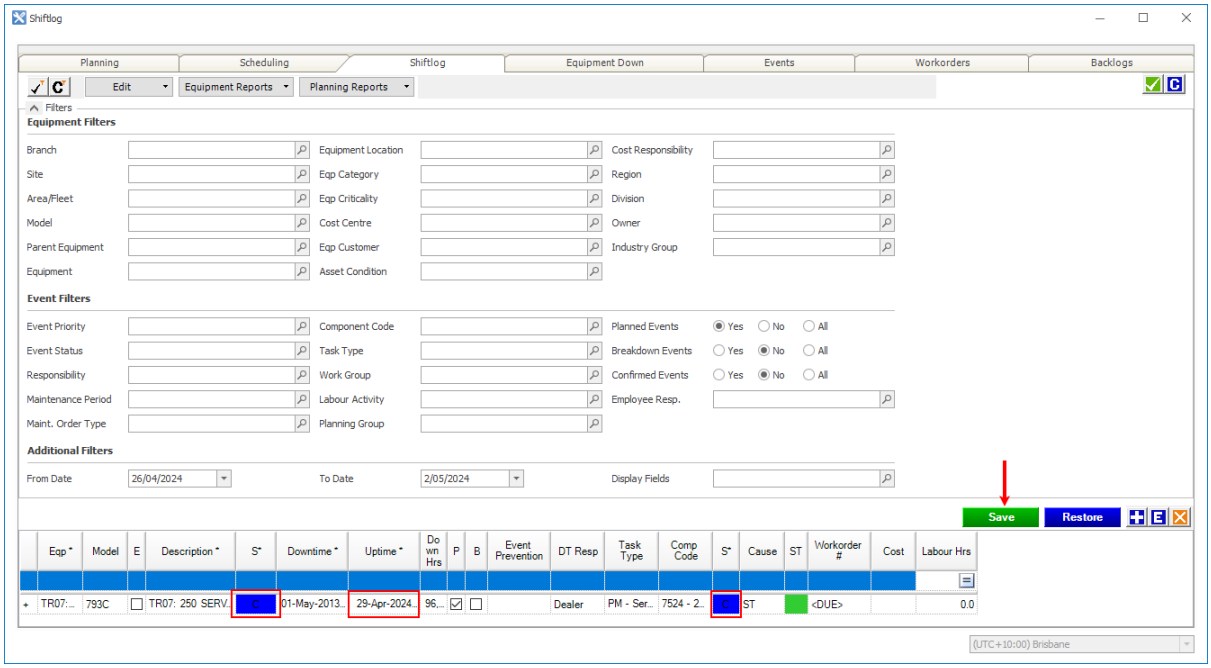
26. Set AMT Breakdown Event to Completed

Procedure – Set AMT Breakdown Event to Completed

If AMT receives a downtime notification from an external dispatch system (such as MineStar, Jigsaw, Modular Mining), that system sends the following information to AMT when the event is completed:

1. Set **Event Status** to **Completed**.
2. Set the notification's **Uptime**.
3. Set the **Workorder Status** to **Completed** (for all work orders linked to the event).
4. Click **Save**.

If AMT is being updated manually, the supervisor will update the existing planned event in AMT using **Shiftlog**.



Exercise – Set AMT Breakdown Event to Completed

1. Log the fact that the asset breakdown event is completed (that is, the asset is back up and running).

27. Complete Job Folder

The mechanic will complete the following documents:

- **Breakdown Workscope** – record the downtime and repair details.
- **Labour** – depending on the processes followed, the mechanics may complete timesheets or enter labour details directly onto the task workscopes.
- **Parts return advice** – details of parts to be returned to stores.

Upon returning to the workshop, all documentation (including defect capture forms from above) is given to the supervisor.

28. Validate Documentation

The supervisor reviews the documentation supplied by the mechanics and checks that it is complete and accurate. If it is not complete or is inaccurate, the supervisor gives the documentation back to the mechanic to complete properly.

Training Workshop (External Downtime)



If this Downtime Interface is not in place, go to the Training Workshop (Work Finalisation).

All asset stoppages (downtime) logged by an external downtime system (such as Modular Mining, Wenco) can be transmitted through a downtime interface to AMT. This section details the steps required to manage the downtime information coming into AMT from this interface.

Recording and reconciliation of asset downtime has been designed to help users to:

- Improve customer relations by ensuring asset downtime is accurately allocated between the various downtime responsibilities.
- Provide opportunities to the daily operations group to include backlog tasks during asset downtime.
- Give the daily operations group better visibility of what is happening NOW to the asset fleet and provide data for understanding WHY things have happened to help prevent future occurrences.
- Provide management with better analysis and reporting of the causes and effects of downtime.
- Achieve cost savings by reducing downtime caused by breakdowns and increasing the level of planned downtime.

29. System Configurations and Settings

By default, external downtime notifications do not automatically create AMT events. However, you can configure AMT to automatically create events based on external downtime notifications.

Furthermore, it is possible for AMT to automatically concatenate downtime notifications into one event. For example, some external downtime systems will resend down and uptime notifications to AMT at the end of a shift. The asset is still down; this is just a quirk of some external downtime systems. If the interfaced data for this occurrence does not send a continuous flow of downtime (that is, the update/time of the previous shift is the same as the down date/time of the next shift), AMT will create two separate events. Even though the asset is still down, and AMT is receiving multiple downtime and uptime notifications for the same asset, AMT will consider these as separate events. However, there is an AMT database setting to get around this issue called **ImportEventMergeMinutes**. This setting can tell AMT to create notifications to be the same event if the notifications occur within *x* seconds for the same asset and, therefore, create one continuous event. This will be agreed upon at the time of implementing the interface.

30. Downtime Interface Process

Functionality Overview

An external downtime system is an external system used for managing asset operations. Downtime events from external downtime (or other production systems) can be interfaced with AMT. This provides accurate data, in real-time, on the downtime and uptime of the assets.

With an interface to these systems, downtime events are automatically created in AMT that match the downtime in the production system.

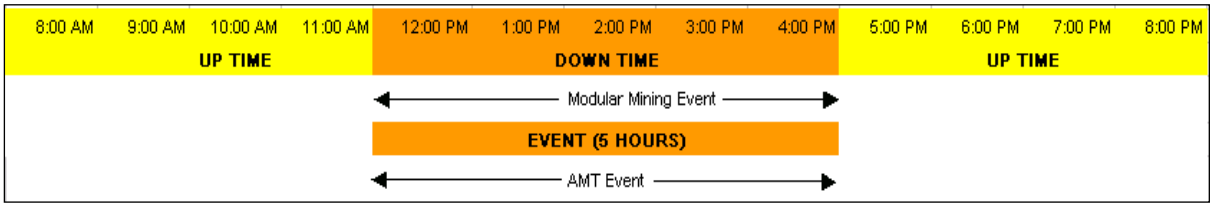
EXTERNAL DOWNTIME TO AMT INFORMATION TRANSFER PROCESS

Following is a generic description of how an external downtime system's downtime notification is generated, transmitted to AMT using an interface, and registered in AMT:

1. On site, the operator notifies of external downtime when an asset stops. This is the start of a downtime event.
2. This event is recorded in the external downtime system and sent to AMT using the interface.
3. AMT receives the downtime notification.
    a. The AMT server receives the new downtime notification from the external downtime system, reads the data, and then logs the notification in the AMT Equipment Status module as an external downtime notification.
    b. The user then has to go into Equipment Status and register the event as an AMT event (this process is explained in a later section).
4. When the asset begins to operate again, the operator logs the asset as back up (that is, it has resumed operation).
5. The external downtime system sends a notification to AMT of the uptime using the interface.
6. AMT appends this notification to the previously generated MM event.

    Now, the external downtime notification in AMT has a downtime and an uptime.
7. This concludes the automated interface process.

The difference between the start and the end of the external downtime notification is the amount of time the asset was down (that is, the downtime). Below is a diagram illustrating how AMT has one AMT event for each external downtime notification (using Modular Mining as an example of an external downtime system).



Linking to the external downtime notification ensures the total downtime for a particular day balances between the two systems. In the above example, both AMT and external downtime will report a total downtime of 5 hours. However, the identification of who is responsible for which portions of the total downtime is not provided as a result of linking. In other words, the linking does not nominate that "maintenance" was responsible for 3 hours of downtime and "production" for 2 hours. It only knows there were 5 hours of total downtime.

The allocation of downtime responsibility between different parties must still be manually entered into the AMT event by editing the event's downtime allocation information.

INFORMATION TRANSFERRED

The external downtime system **DOES** provide AMT with:

- The identity of the asset that stopped.
- The time the asset stopped (actual downtime).
- A reason for stopping (a cause code and description).
- The time the asset resumed normal operation (actual uptime).

The external downtime system **DOES NOT** provide AMT with:

- Who was responsible for the downtime (dealer, customer, opportunity)?
- Coding for tasks performed (component code, modifier code, task type).
- Additional tasks performed (the linking process only creates 1 task).
- Work orders.
- Symptom and cause of the stoppage.
- The repair performed, and so on.

After accepting the external downtime notification (that is, making it an AMT event), the work order created in the event is set to status Complete. The AMT event needs to be reviewed and, work orders that are not completed need to be deferred, deleted or abandoned. The user must supply missing work orders and missing downtime information to the AMT event (as outlined in the previous section).

Remember, the quality of reporting from AMT depends on the quality of data in the system. Equipment status is the cornerstone of data integrity in AMT. This is the area where all the work gets done.

31. Managing Downtime Notifications

AMT can be configured to automatically create AMT events from notifications54 or to place them in a holding area, and you can select which notifications will be used to create actual downtime events in AMT. Each approach is explained below.

MANUALLY CREATE EVENTS

If AMT is configured in this manner, external downtime notifications will be placed in a holding area within AMT. This holding area is accessible using the External Downtime wizard. Using this wizard, you can see all downtime notifications received by AMT and then determine which of the notifications will be:

- Ignored (this does not delete the notification).
- Linked to an existing AMT downtime event (such as a planned event), in which case, search for and select the AMT event to link the notification to.
- Used to create a new AMT event (such as a breakdown event).

AUTOMATICALLY CREATE EVENTS

If AMT is configured in this manner, all external downtime notifications will appear as AMT events in Planning and Equipment Status. You need to:

- Delete the external event if it is not required.
- If the external event represents planned work, merge the external event with an existing AMT planned event. To do this, open the new external event, click the **Merge Yet to Start Event** function, and select the AMT YTS event.
- If the external event represents breakdown work, update the external event's downtime allocation and work order details.

REMOVING EXTERNAL DOWNTIME EVENTS

There are two ways that you can remove an external downtime event:

- Unlink the internal AMT event from the external downtime event. This disassociates the two events. However, the external downtime event will remain visible in Equipment Status.
- Delete the external event like you delete an internal AMT event. However, once an external downtime event is deleted (from Equipment Status) it remains visible in the External Downtime wizard. You can choose to ignore the event so that another user does not manually try to recreate it again.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

Training Workshop (Work Finalisation)

32. Perform Planned and Breakdown Work

After planned and breakdown work is completed, the job folder documentation signed off by the supervisor is used to complete the **Work Finalisation** process.

For more information about the processes for **Perform Planned Work** and **Perform Breakdown Work**, see earlier sections in this document.

33. Enter New Defects into AMT

Defects documented, but not actually performed during the **Perform Work** processes (such as work orders where parts need to be ordered), are entered into AMT by the mechanic. These defects will appear as backlogs in AMT, which a planner will schedule at a future time.

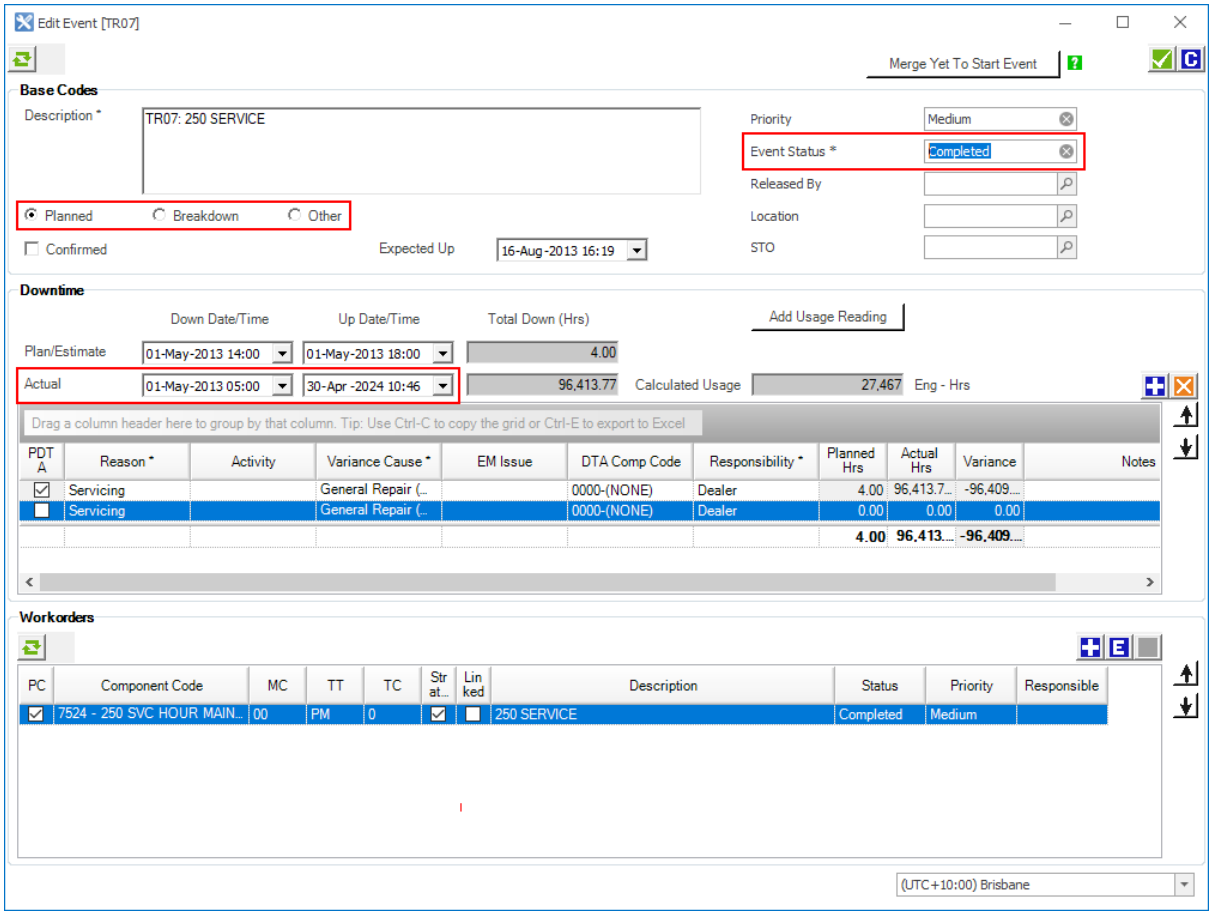
34. Update AMT with Downtime and Feedback Information

Procedure – Update Event

1. Open the **Daily Operations** > **Equipment Status** > **Shiftlog** tab, and set **Event Status** to **Completed**.
2. Filter for the event.
3. Click **Edit**.



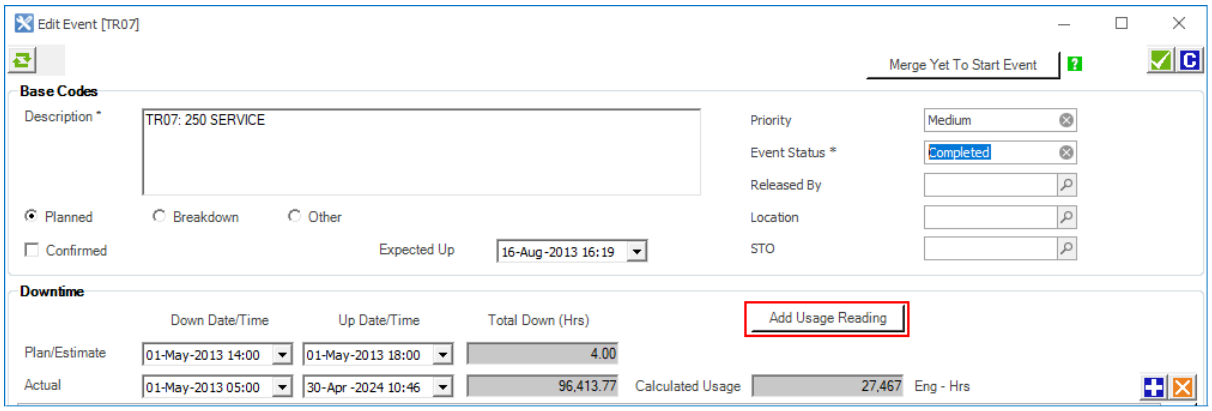
This update will either be performed manually or automatically if there is an interface to a dispatch system.



4. Enter the **Base Codes** (Breakdown, Planned, Other). In the case of Breakdown, AMT will ask you to enter the Event Prevention option.
5. Enter the **Actual Downtime** and **Uptime**.
6. Change the **Event Status** to **Completed**.

Procedure – Usage Reading

1. Click **Add Usage Reading** to update the **Actual Usage** table with the latest usage reading of the asset as of the start of the downtime.



2. Enter the usage reading and click **Save**.

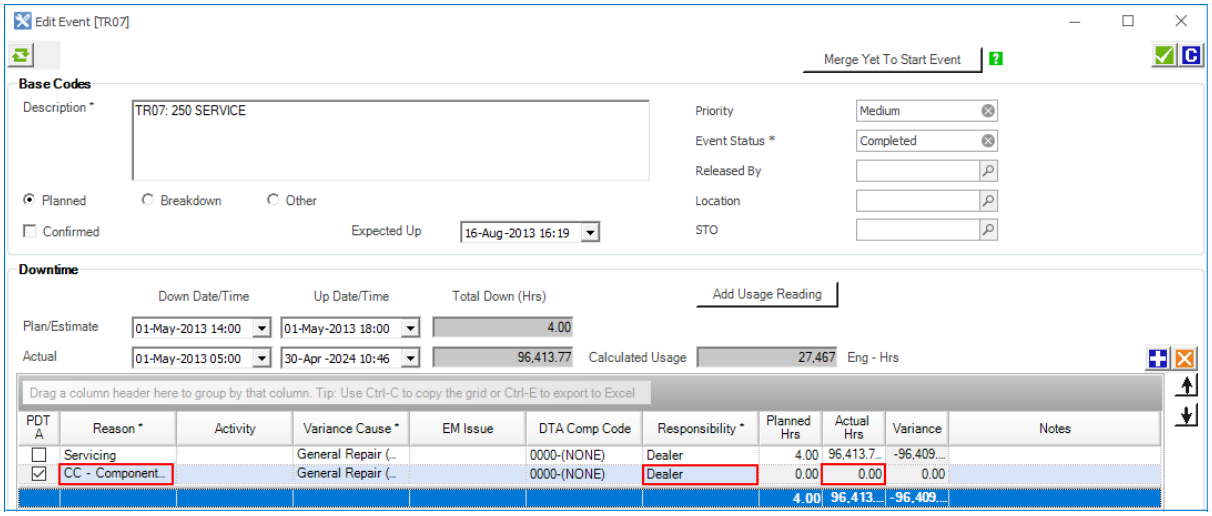
Procedure – Downtime Allocation

1. On the **Shiftlog** tab, click **Edit**.



*If there is split responsibility or delays, you can **Edit** the event and enter multiple downtime allocations.*

2. Adjust the downtime allocations by adding a line.
3. Add a **Reason** for the additional downtime.
4. Set the **Responsibility**.
5. Update **Actual Hours**.



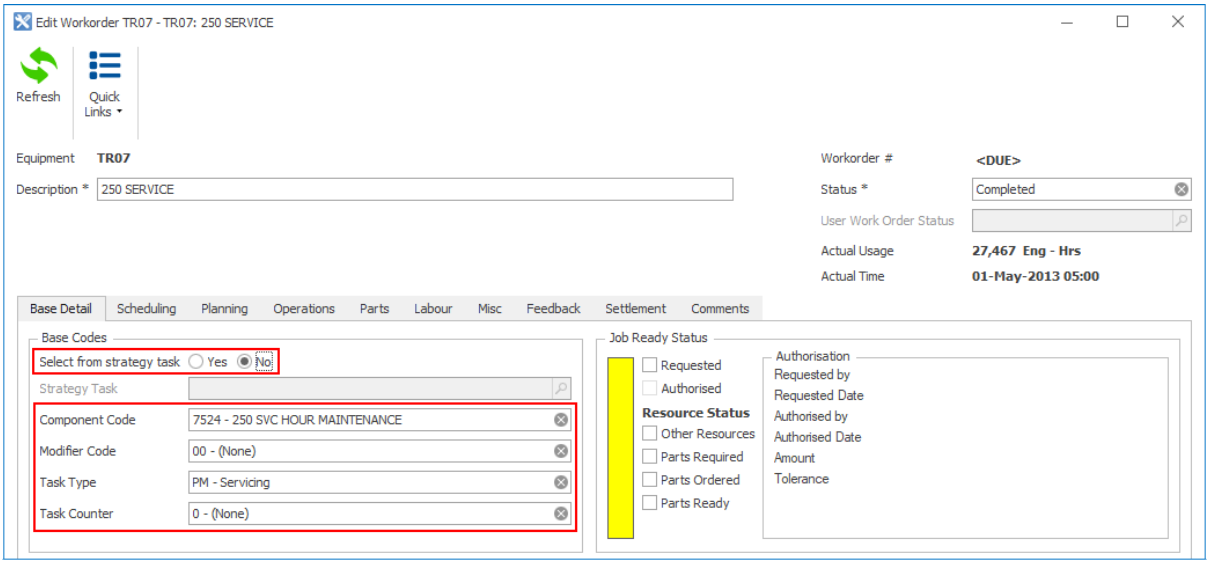
Procedure – Work Orders

- **Status = Completed**: For the work orders that are completed.
- **Status = Yet to Start**: Event not yet started, but still planned to be done.
- **Status = In Progress**: Where the work order is still in progress.
- **Status = Deferred**: for the work orders not performed. This will re-instate the work order in the Planning grid with a status of **Outstanding** (click **Edit** to open the event and set the work order status to **Deferred**).
- Work order codes (**Task Type**, **Component Code** and **Cause**).

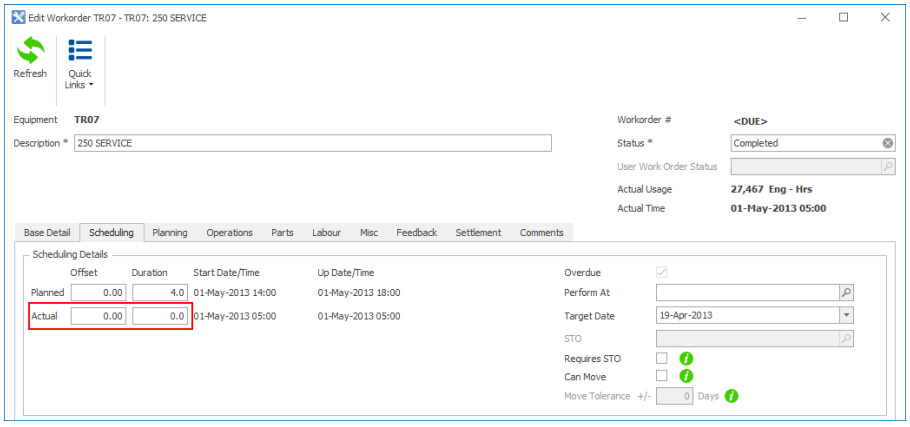
Additional coding and notes can be added on the **Feedback** tab of the work order.

This step focuses on the information provided by the mechanic about the work that was performed, failure codes for analyses, technical information on components and warranty claim details. Depending on the configuration, you can also add actual parts usage information and actual labour hours (these will only be entered if not being imported from an ERP system). The coding should be correct.

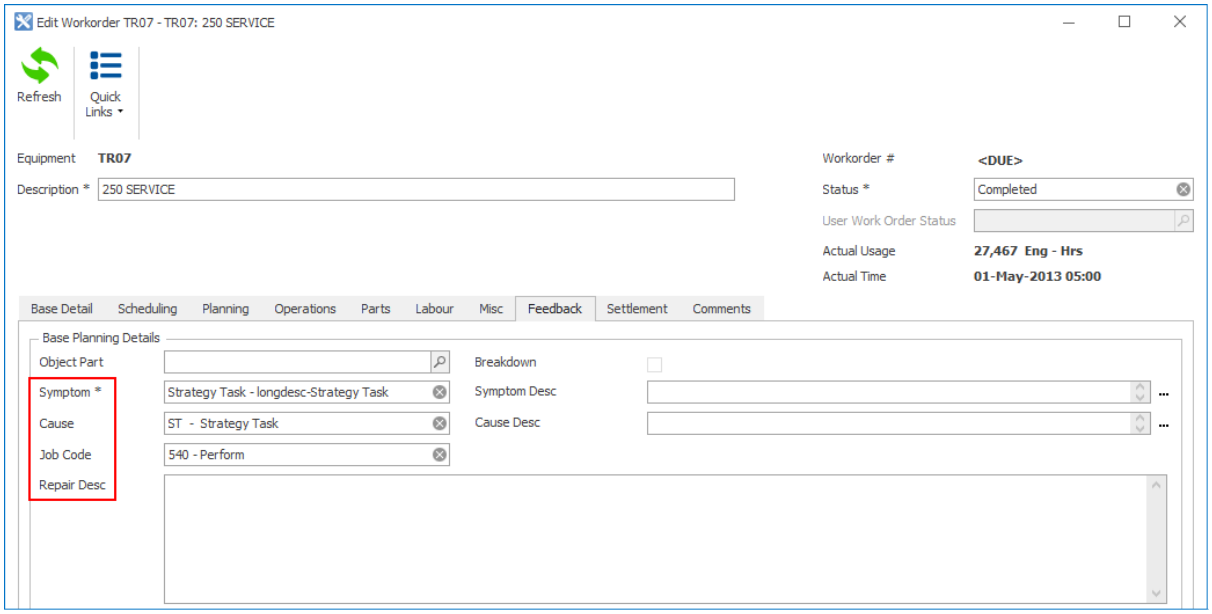
1. Right-click the work order and select **Edit Workorder**.
2. On the **Base Details** tab, determine if the Workorder should have been a **Strategy Task**.
3. Confirm the **Component Code**, **Modifier Code**, **Task Type** and **Task Counter**.



4. On the **Scheduling** tab, update the **Actual** work order duration.



5. On the **Feedback** tab, update the work order feedback details from the Workscope: **Symptom**, **Cause**, **Job Code** and **Repair Description**.



Exercise – Update AMT with Downtime Feedback Information

1. Record event details.
2. Record 2 downtime allocations (assume that the last hour of downtime was “waiting for operator”).
3. Record work order details (such as status, codes, fields on the Feedback tab).
4. Update the event actual usage.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

35. Provide Job Folders in Asset Files

If required, send hard-copy documents to the Planner to review and file.

36. Record Additional Work Order Details

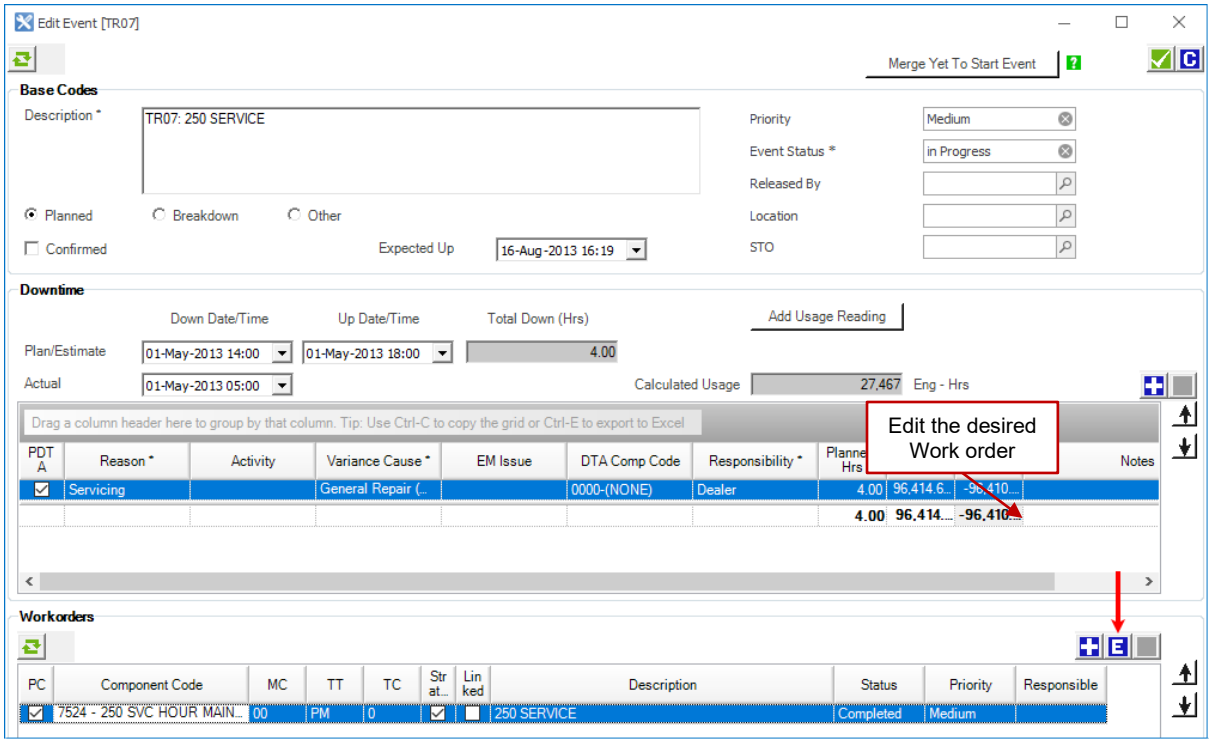
The planner should review the event and work order details to understand what happened during planned events that they scheduled, as well as understand what else has happened to the fleet. Use the Shiftlog to view and edit completed events and work orders.

From the review, the planner may identify:

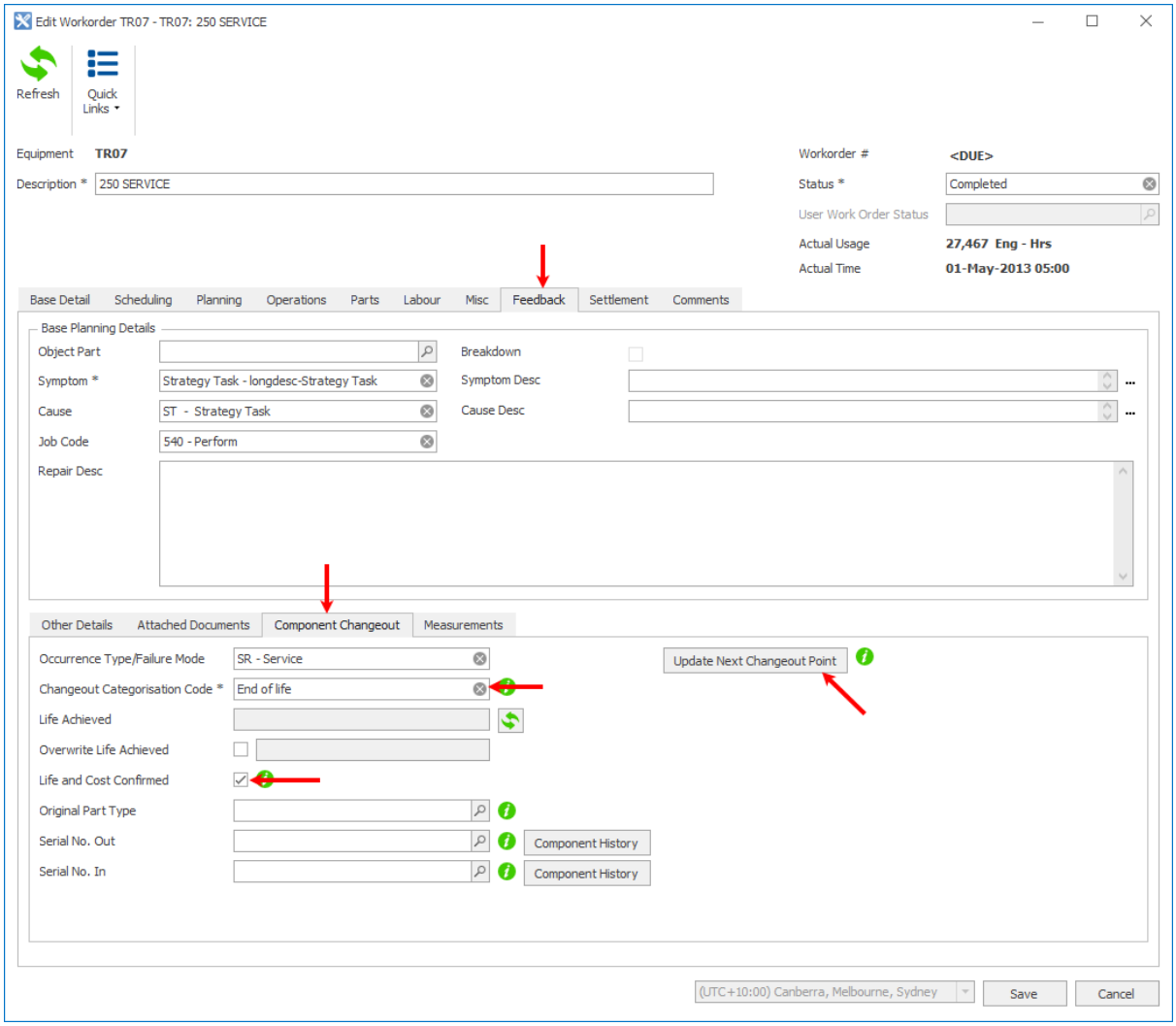
- Missing costs or costs allocated incorrectly.
- Accidental errors made by supervisors or mechanics in updating AMT.
- **Swing Component** movements that need to be flagged in AMT.

Procedure – Update Swing In and Swing Out Component Details

1. On the **Shiftlog** screen, use filters to locate the specific event.
2. Click **Edit**.
3. Click **Edit Workorder**.



4. Click the **Feedback** tab, then click the **Component Changeout** tab.



5. If the major component is being removed for repairs, resulting in a temporary substitute component being installed, set the following values (as shown in the screenshot above):

   a. **Changeout Categorisation Code** = Swing In.

      This tells AMT that the component going into the asset is temporary.

   b. **Life and Cost Confirmed** = Checked.

      This tells AMT that this work order is a component changeout and that the life and cost achieved by this component are confirmed as correct.

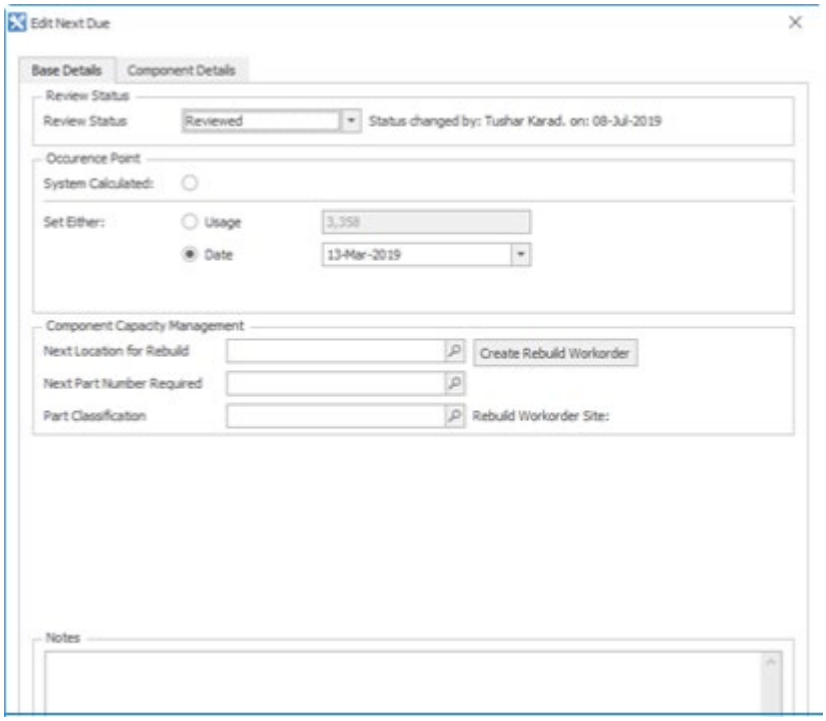
Record when the component being repaired will be reinstalled, in order to ensure AMT's scheduling is correct. To do this:

1. Click **Update Next Changeout Point** (shown above), and set either the **Date** or **Usage** when the repaired component will be reinstalled.

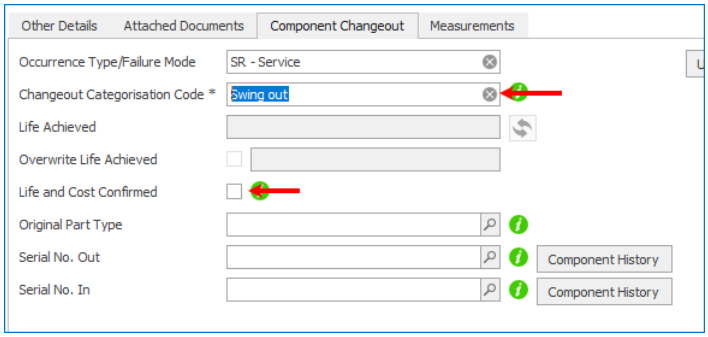


This button is only enabled after the work order is in a status of In Progress or Completed, and is linked to a work order settlement.

2. The strategy task will open, click **Edit Next Due** under **Quick Links**.
3. Select either the **Next Usage** or **Next Date** to reflect when the repaired component will be reinstalled.
4. Enter a **Changeout Reason** (such as reinstall repaired component and remove swing component).
5. Set **Review Status** = Reviewed.
6. Click **Save and Close**.

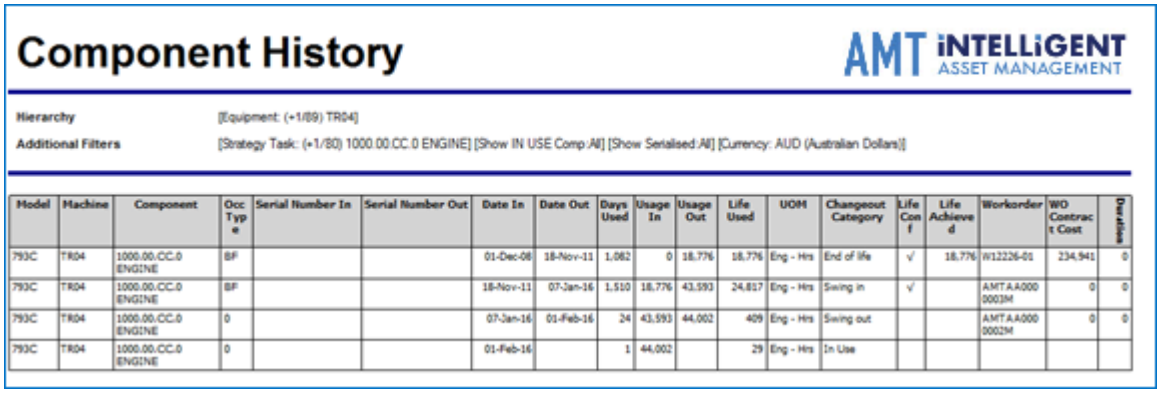


7. When it is time to reinstall the repaired component (and therefore also remove the swing component), create the event and the work order for the same strategy task. However, this time the settings are different:
   a. **Changeout Categorisation Code** = Swing Out.
      This tells AMT that the temporary component is being removed.
   b. **Life and Cost Confirmed** = Unselected.
      This tells AMT that the life achieved by the component being removed should not be reported as full life because it was a swing component.



8. Click **Save** (you do not need to update the next changeout point because AMT will automatically reset the life).

9. The results of this process can be seen in the **Component History** report (**Strategy Optimisation** > **Component Reliability Analysis**).



Component History

AMT INTELLIGENT ASSET MANAGEMENT

Hierarchy [Equipment: (+1/89) TR04]

Additional Filters [Strategy Task: (+1/80) 1000.00.CC.0 ENGINE] [Show IN USE Comp:All] [Show Serialised:All] [Currency: AUD (Australian Dollars)]

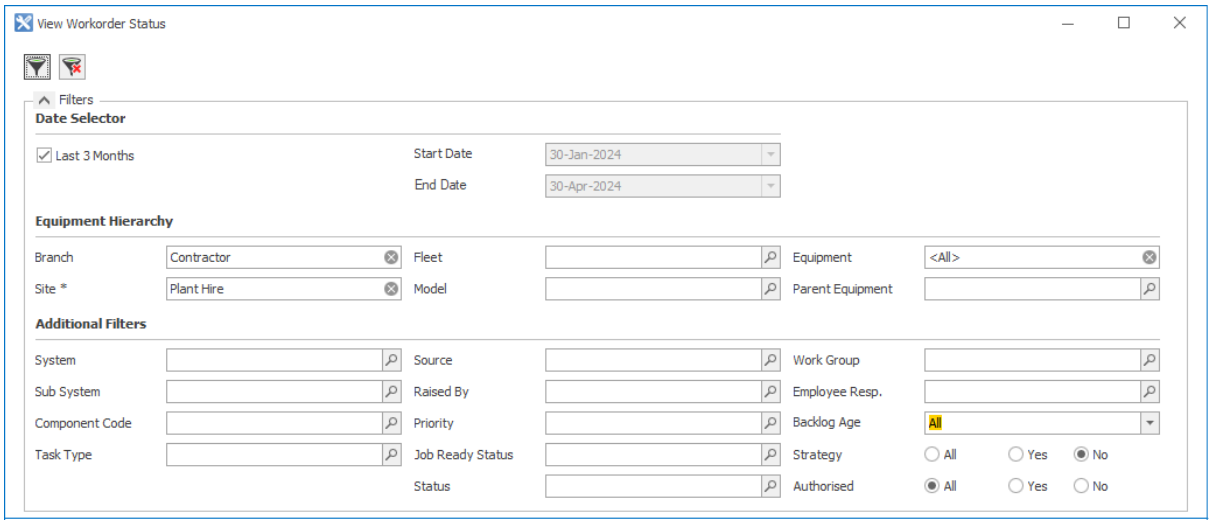
Model	Machine	Component	Occ Type	Serial Number In	Serial Number Out	Date In	Date Out	Days Used	Usage In	Usage Out	Life Used	UOM	Changeout Category	Life Conf	Life Achieved	Workorder	WO Contract Cost	Duration
793C	TR04	1000.00.CC.0 ENGINE	BF			01-Dec-08	18-Nov-11	1,082	0	18,776	18,776	Eng - Hrs	End of life	√	18,776	W12226-01	234,941	0
793C	TR04	1000.00.CC.0 ENGINE	BF			18-Nov-11	07-Jan-16	1,510	18,776	43,593	24,817	Eng - Hrs	Swing in	√		AMT.AA000 0001M	0	0
793C	TR04	1000.00.CC.0 ENGINE	0			07-Jan-16	01-Feb-16	24	43,593	44,002	409	Eng - Hrs	Swing out			AMT.AA000 0002M	0	0
793C	TR04	1000.00.CC.0 ENGINE	0			01-Feb-16		1	44,002		29	Eng - Hrs	In Use					

Exercise – Record Swing In / Swing Out Component

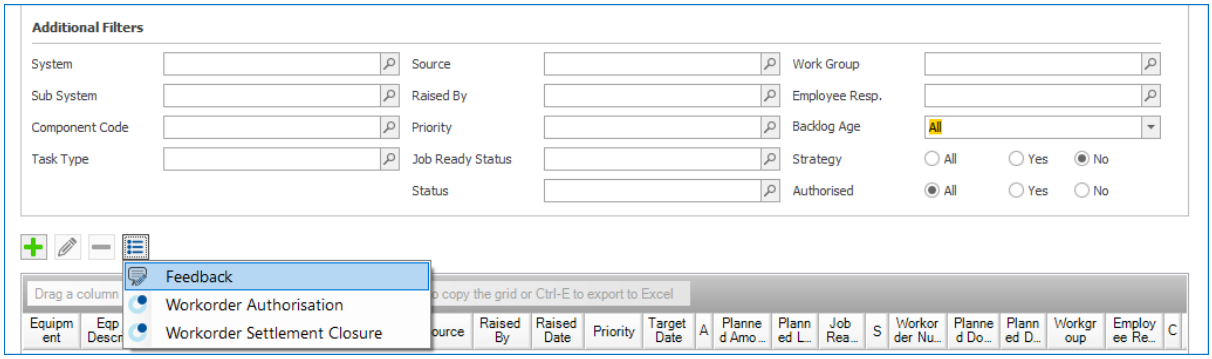
1. Select a major component of an asset (such as the engine of a haul truck).
2. Record the Swing In and Swing Out work orders for the major component.
3. View the results in the Component History report.

Procedure – Enter Component Tracking Details

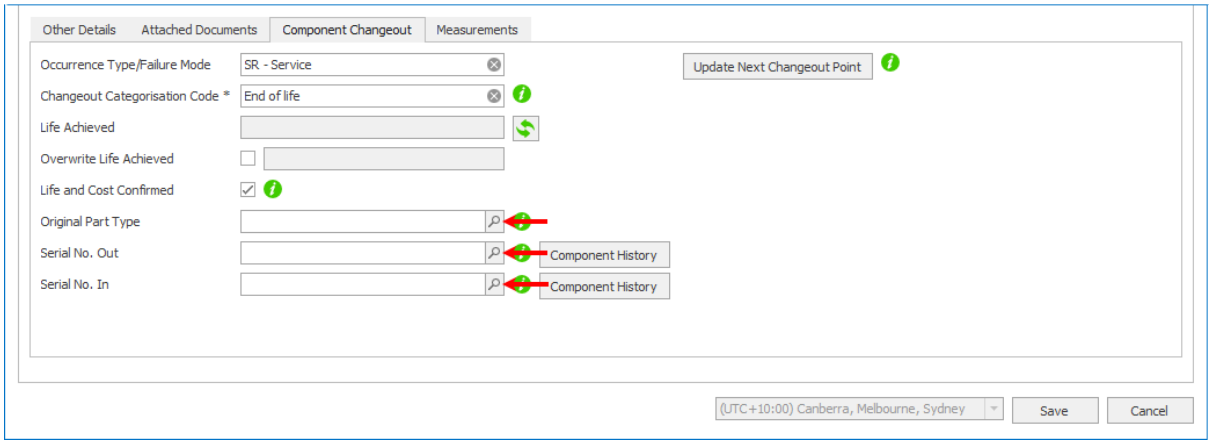
1. Under **Daily Operations**, double-click **View Workorder Status**.
2. Enter the filter criteria and click **Apply Filter**.



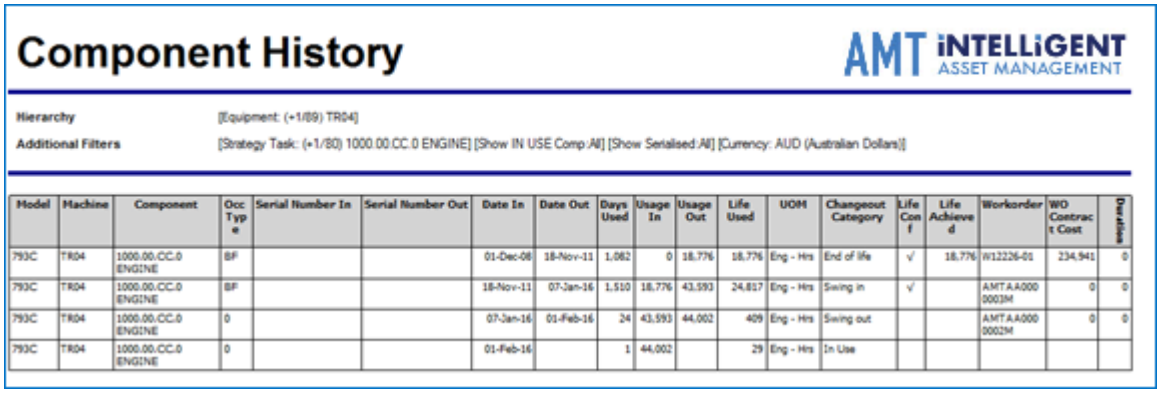
3. Highlight the workorder, click the **Quick Links** button, then click **Feedback**.



4. Click the **Component Changeout** tab.
5. Enter the serial number of the component that was removed in the **Serial No. Out** field.
6. Enter the serial number of the component that was installed in the **Serial No. In** field.
7. Enter the details of the component serial numbers.



8. Click **Save and Close** when finished.
9. Run the **Component History** report to see the serial number data.



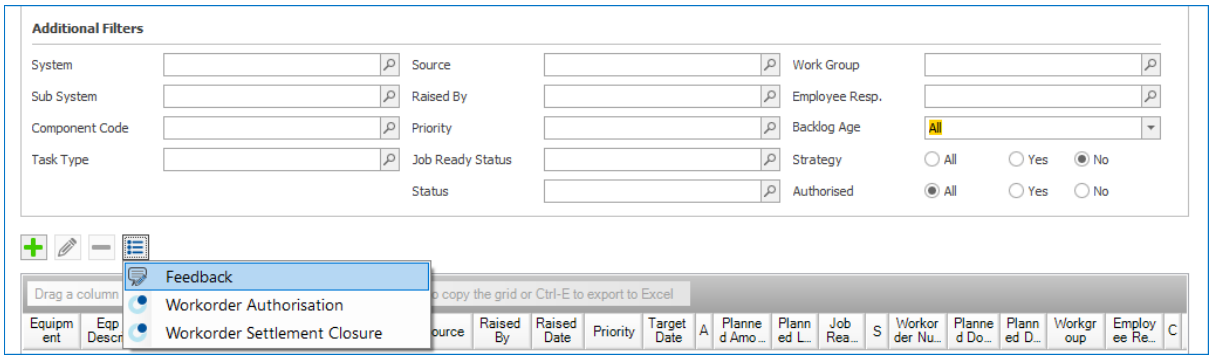
10. To see the history of a particular serial number, set the report filters.
    - **Component Serial Number** = desired serial number and run the report.
    - **Show IN USE Comp** = ALL.
11. The report filter **Exact Serial Number Match** should be flagged if you want to look for a specific serial number entered in the **Component Serial Number** field.

Exercise – Record Component Tracking Data

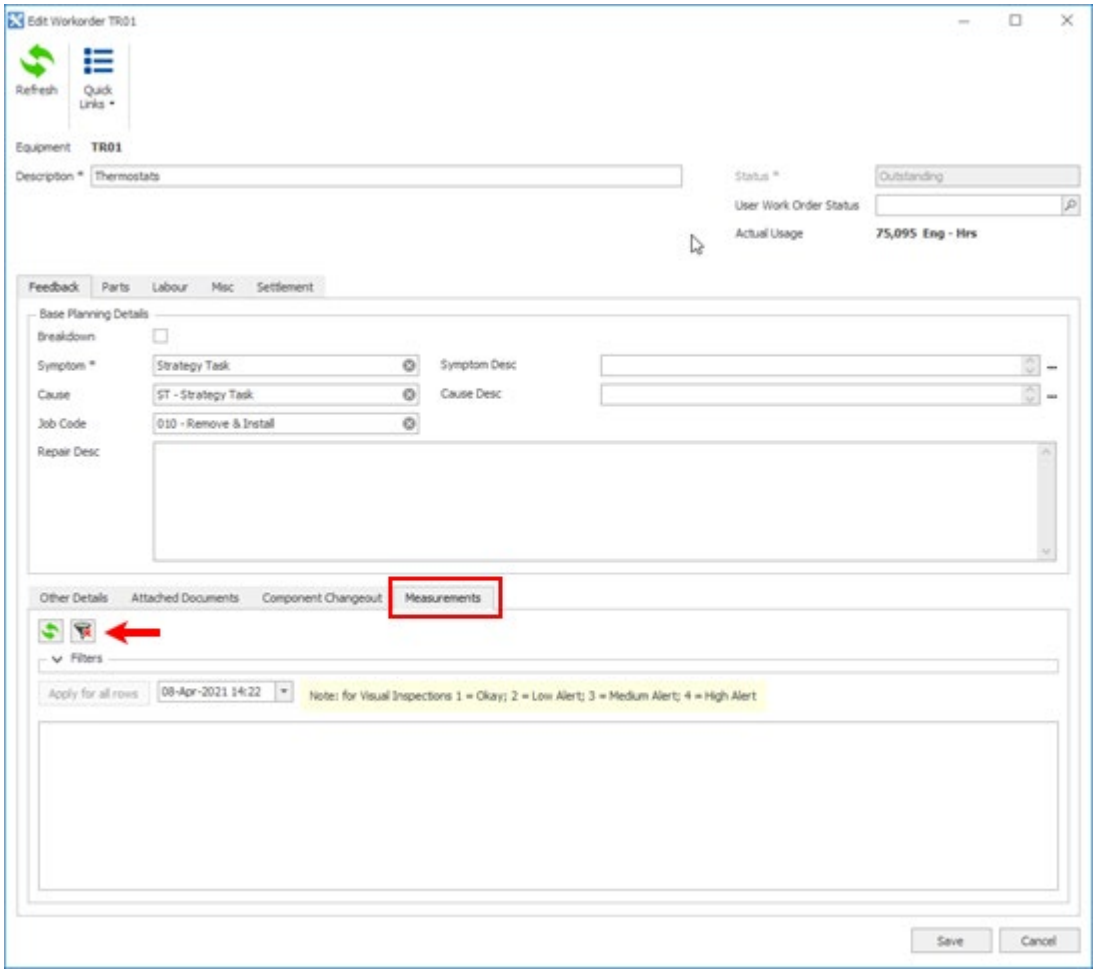
1. Select a major component (such as the engine of a specific model of haul truck).
2. Register the **Serialised Component** in AMT.
3. Record several work orders on different assets to represent the historical lives of the serialised component.
4. Verify the history of the component by running the **Component History** report.

Procedure – Enter Measurements on a Work Order

1. Under **Daily Operations**, double-click **View Workorder Status**.
2. This provides a daily snapshot of work order status, so you can provide quick feedback updates.
3. Enter the filter criteria and click **Apply Filter**.
4. Click the **Quick Links** button, then click **Feedback**.

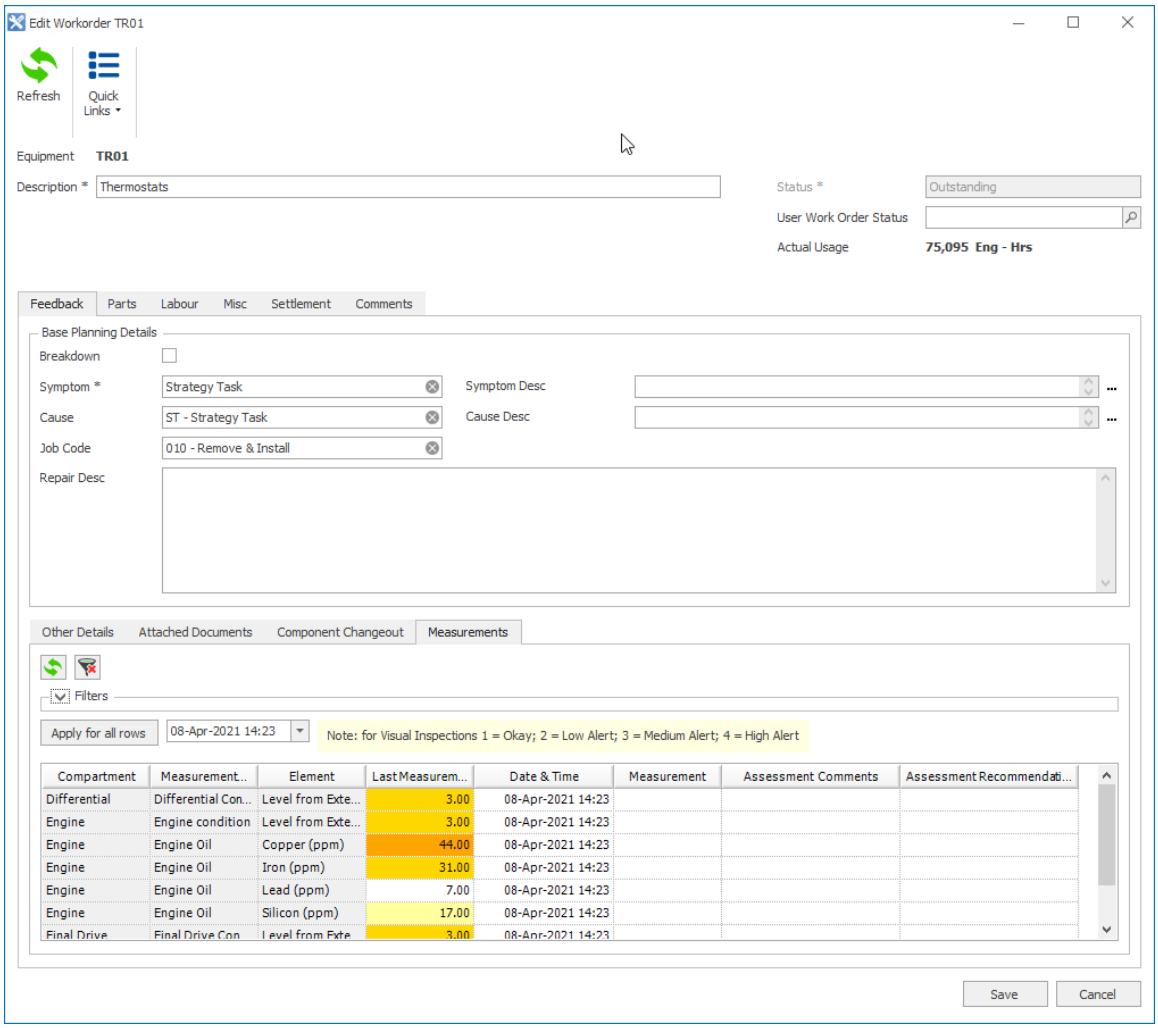


5. Click the **Measurements** tab.
6. Click **Apply Filters** to show application measurements.
7.



AMT will show the value and date of the last measurement captured in AMT.

8. Enter the new measurement value in the **Measurement** column. You can also change the **Date and Time**, enter **Comments** and/or a **Recommendation** (such as a suggested follow-up action).
9. By default, all measurements flagged as being able to be captured on a work order for this model of the asset will be displayed.
10. If required, enter values for other types of measurements (the measurements are limited to those applicable to the model of the asset).



11. Click **Save and Close**.

Exercise – Record Measurements in a Work Order

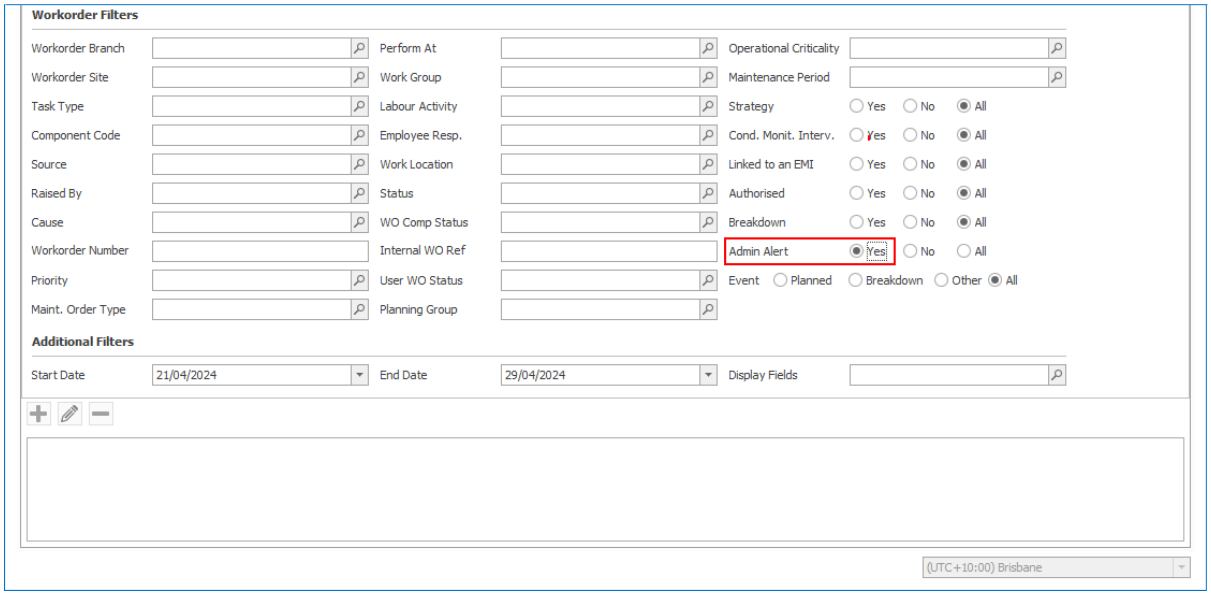
1. Find a work order and record measurement readings (You may need an administrator to set up valid measurement codes first).

Procedure – Admin Alert

A work order will be flagged with an **Admin Alert** if one of the following conditions is met:

- It is a **strategy work order**, but is not linked to a work order settlement.
- It is a **rotable work order**, where rotable records have not been created.
- The event is **COMP**, but the work order is IP or YTS.
- The event is **YTS**, but the work order is IP or COMP.

Admin Alert is an available filter in the **Workorders** tab, to help identify work orders that need attention.



Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

37. File Job Folders in Asset Files

If required, the planner should file Job folder documents in asset files for audit and company policy purposes.

38. Reconcile Downtime to Dispatch

The planner will then reconcile AMT downtime to dispatch. This process includes ensuring the allocation of downtime between operations and maintenance is accurate.

Use the following AMT reports for this process (remember to set the Responsibility filter):

- Daily Downtime Log.
- Equipment Operational Performance Summary.
- Daily Machine Performance Check Sheet.

The planner should make updates using **Equipment Status** or **Shiftlog**.

Exercise – Run and Understand Daily Downtime Reports

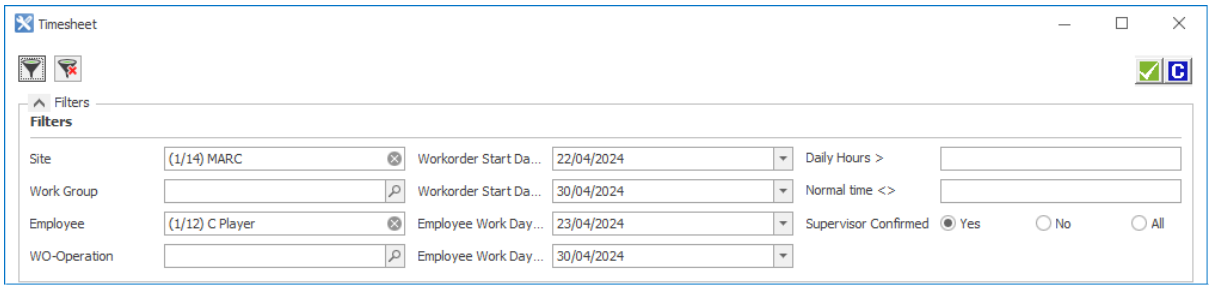
1. Run the reports covered in this section.
2. Validate the reports match downtime entered into AMT for a one week period.

39. Enter Employee Timesheets

Procedure – Enter Employee Timesheets

The Timesheet function in AMT is a simple solution to review and capture the actual hours worked by employees.

1. Under **Daily Operations**, double-click **Timesheet**.
2. Select **Filters**, **Workorder Start Date Range**, and **Employee Work Days**, then click **Apply Filters**.



We recommend that you enter employee work time by either adding all work time for one employee at a time or by entering data for all employees on the roster for a single day.

3. If an employee worked on more than one operation, click **Add** to create a separate line. Only work orders for the date selected appear in the Workorder-Operation filter.
4. Click **Save**.

Exercise – Enter time in the Timesheets

1. Enter actual labour hours for 3 employees for a particular day.

40. Produce a Daily Log Report for the Daily Production Meeting

On many sites, there is a daily meeting between maintenance (planners and supervisors) and operations to review the events of the previous day and agree on the plan for the current day. This is often complemented by a more formal weekly meeting.

A number of daily log reports exist in AMT to summarise the events of the previous day, including:

- Daily Downtime Log
- Shift Downtime Log
- Daily Log – Detailed
- Daily Log – Summary
- Monthly Downtime Log

Exercise – Run and Understand Daily Log Reports

1. Run the reports covered in this section.
2. Validate the reports match downtime entered into AMT for a one week period.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

41. Review and Confirm Events

Procedure – Review & Confirm Events

Events can be confirmed. This means that you are flagging the event as being reviewed and all data confirmed to be correct. AMT will check certain rules (for example, required fields) when you attempt to confirm an event.

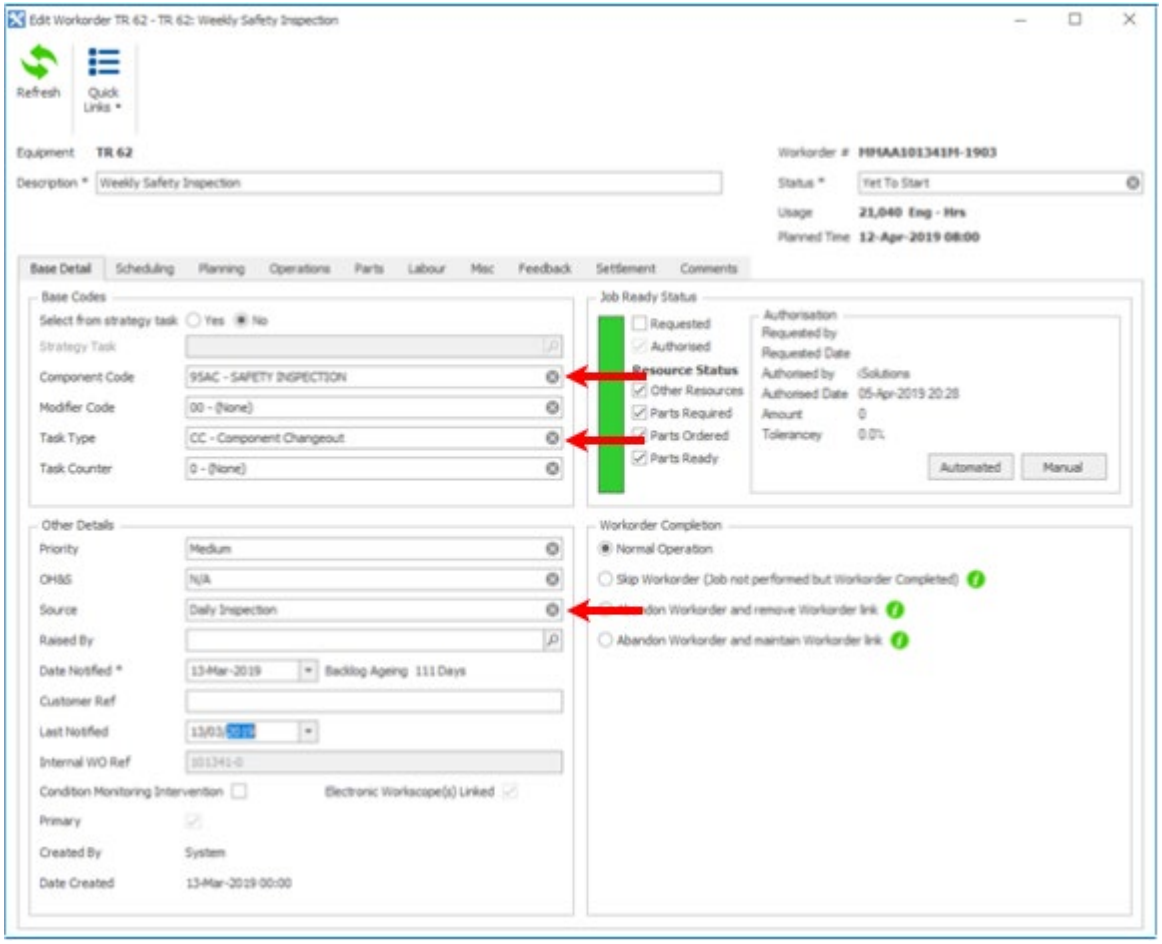
Navigate to **Daily Planning** > **Planning**.

Fields that need to be completed before confirming an event include:

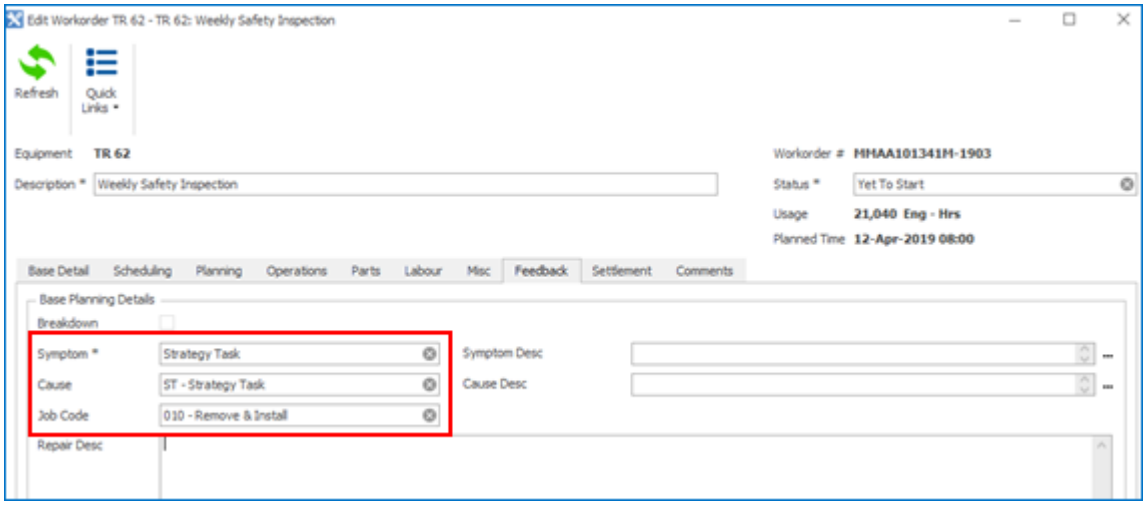
1. **Released By** in the **Event** header.



2. **Component Code**, **Task Type** and **Source** in the work order on the **Base Detail** tab.



3. **Symptom**, **Cause** and **Job Code** in the work order on the **Feedback** tab.



4. After these fields have been entered, save the work order.
5. Click **Save and Close**.

Exercise – Confirm an Event

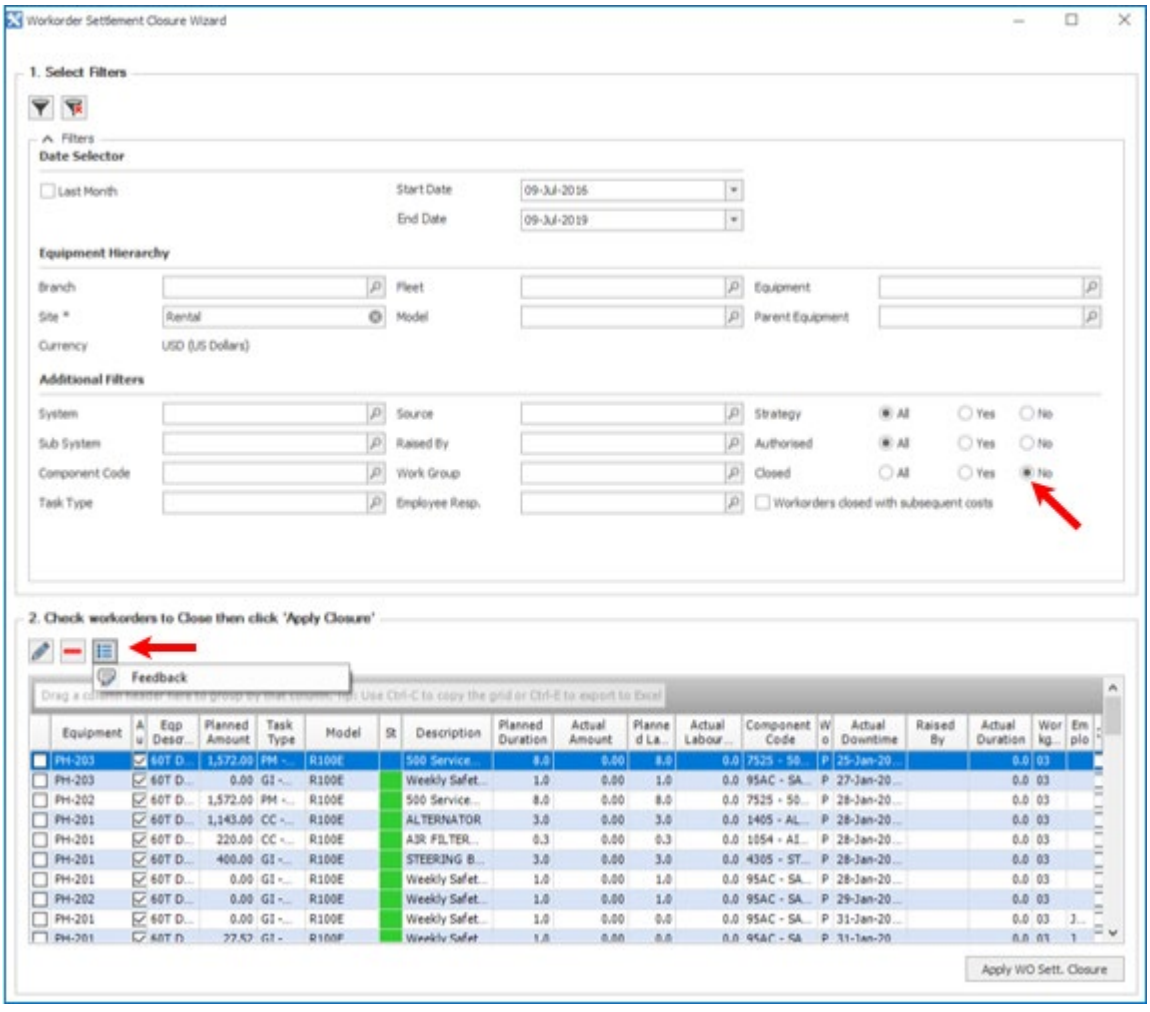
1. Find a completed event and ensure that all of the required fields have been completed.
2. Confirm a completed event.

42. Work Order Settlement Closure

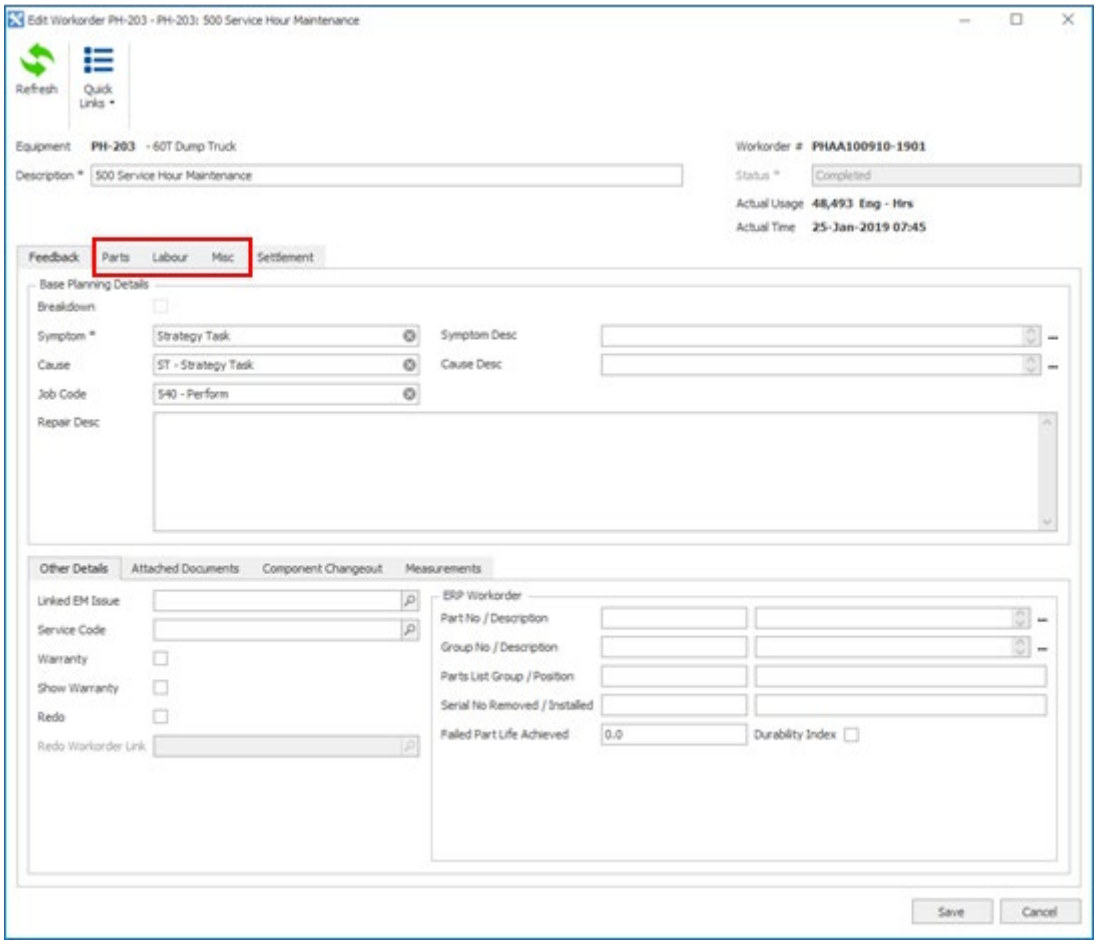
Procedure – Work Order Settlement Closure

If AMT does not interface with an external ERP work order system, the supervisor must also close work order settlements.

1. Under **Daily Operations**, double-click **Workorder Settlement Closure**.
2. Set all of the required fields and click **Apply Filters**.
3. After the correct work order is located, highlight it and click **Feedback**.
4. Since the objective is to close outstanding work orders, make sure **Closed** is set to **No**.



5. Check if the values stated under **Parts**, **Labour** and **Misc** are correct. If not, correct them.

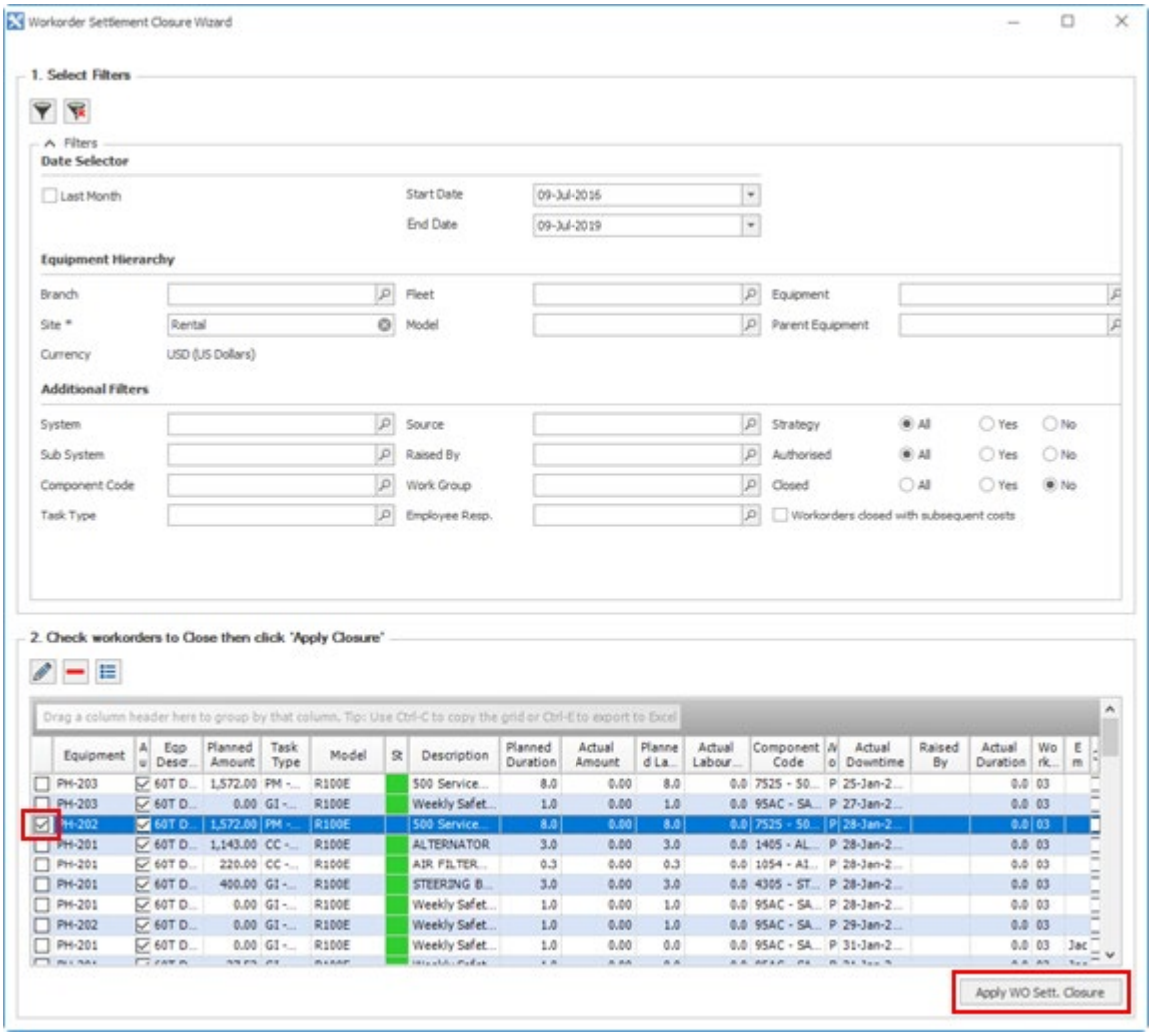


6. If changes have been made to the work order, click **Save**. Otherwise, close the window to return to the previous windows.

7. Select the work orders to be closed by selecting the first check box, then click **Apply WO Set Closure**.



This option is only applicable if AMT is THE ONLY work order system.



You can use this process to close work order settlements for multiple work orders.

8. Use the Workorder Authorisation option to authorise multiple work orders (Refer to **RPM Training – AMT Short Term Planning** manual for **Bulk Workorder Authorisation** steps).

9. AMT will save the date, time and user who applied the closure.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

Training Workshop (Usages, Measurements and Production Statistics)

A key component of the asset management process is capturing vital asset information. This information can range from idle time, operating hours, cycles used, litres consumed, kilometres travelled, to tonnes moved. Collecting this data is not only important from an operational perspective to measure performance, but is equally critical in management of the asset strategy from a maintenance perspective. There are three main areas where with information can be maintained in AMT:

- Actual usages (can be maintained manually or using an interface).
- Measurements (can be maintained manually or using an interface).
- Production statistics.

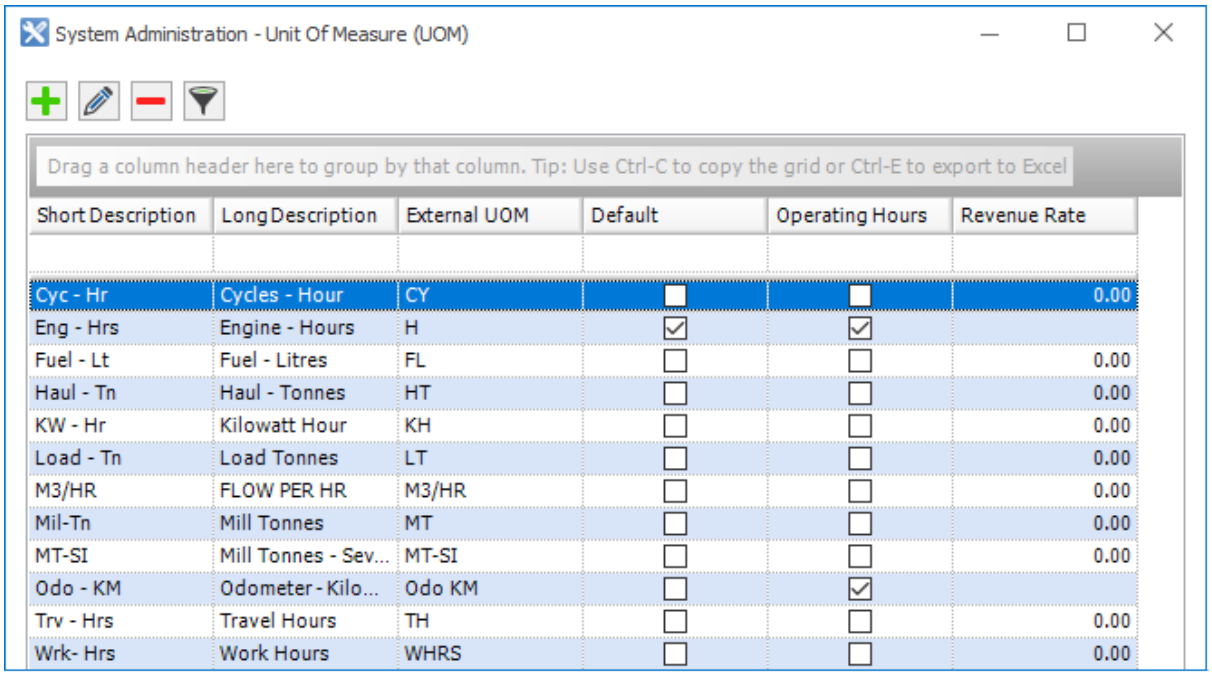
43. Actual Usages

As part of the Work Finalisation process, the actual usages of an asset are typically updated on the Workscopes. This section shows you how to manually update usages directly in the Actual Usage Details.

Procedure – Actual Usages Setup

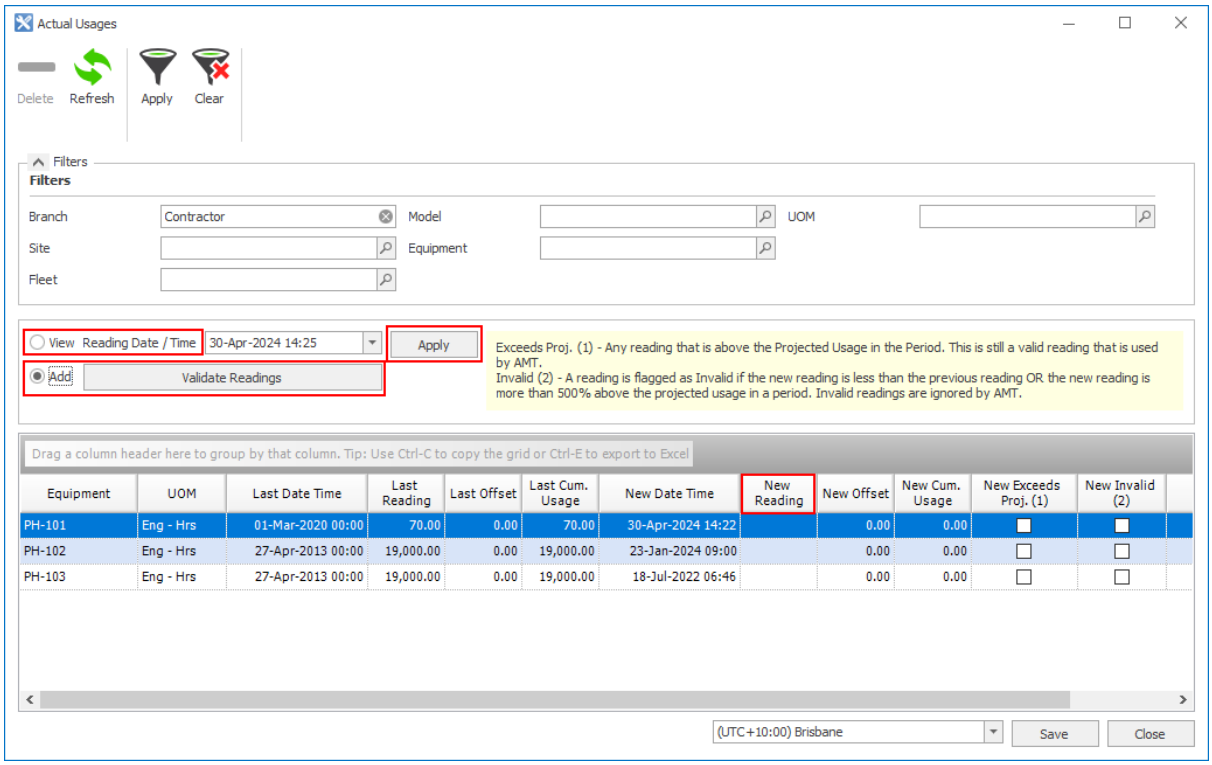
Ensure that the relevant UOM has been setup.

1. Navigate to **System Administration** > **System Tables** > **Usages & Measurements**, double-click **Units of Measure**.
2. Ensure that at least one UOM is selected as **Operating Hours**.



Procedure – Record Actual Uses

1. Navigate to **Daily Operations** > **Usages & Measurements**, double-click **Record Actual Usages**.
2. Filter for the relevant assets and select **Add**.
3. Enter the date and time that the reading was taken, and click **Apply**.
4. Enter the new readings in the **New Reading** column, select **Validate Readings** and make adjustments if anything is invalid.



5. After you've made your updates, click **Save**.

You can right-click on the grid, select and copy the data into a spreadsheet program. In the spreadsheet, you can complete New Date Time, New Reading and/or New Offset value and then paste the full grid back into AMT.

Procedure – Usage Validation

Usage measurements are checked and determined if they are valid:

- **Invalid (scenario 1)** - check if the new reading is lower than the previous reading. AMT does not allow this type of entry for manual readings, but it is possible if the readings are imported.
- **Invalid (scenario 2)** - check if the new reading is more than the Projected Usage in the Period but exceeds the threshold by more than 500%.
- **Exceeds Projected Usage** - check if the new reading is more than the Projected Usage in the Period but no more than the 500% threshold.

Occasionally, the meter reader is changed or damaged on an asset and the counter needs to be reset and started again. When this happens, the offset is set by the user to be equal to the last reading prior to the reset (for example, 34,238 engine hours). The **New Reading** might then be equal to zero.

Procedure – Analyse Usages

Use the Asset Performance reporting area of AMT to interrogate a wide range of performance data to produce a variety of asset performance and asset productivity reports. For utilisation and usage analysis, navigate to **Equipment Performance**.

You can view this data in report format:

1. Actual usages that have been recorded.
2. Expected usages, past and future (based on the asset's) usage profile.
3. Actual versus expected usages.

Exercise Record Actual Usages

1. Record actual usages for 5 assets.
2. Validate the 5 new actual usage readings.
3. Enter a new offset for one of the assets.

44. Measurements

When actual measurements are received by AMT, they are classified based on user-defined translation rules. The rules help you interpret whether the measurement received is an alert or OK (normal). Based on the valid readings received by AMT, you can run a variety of reports to analyse consumption. Measurements are also used for condition-based maintenance analysis if this module is installed.

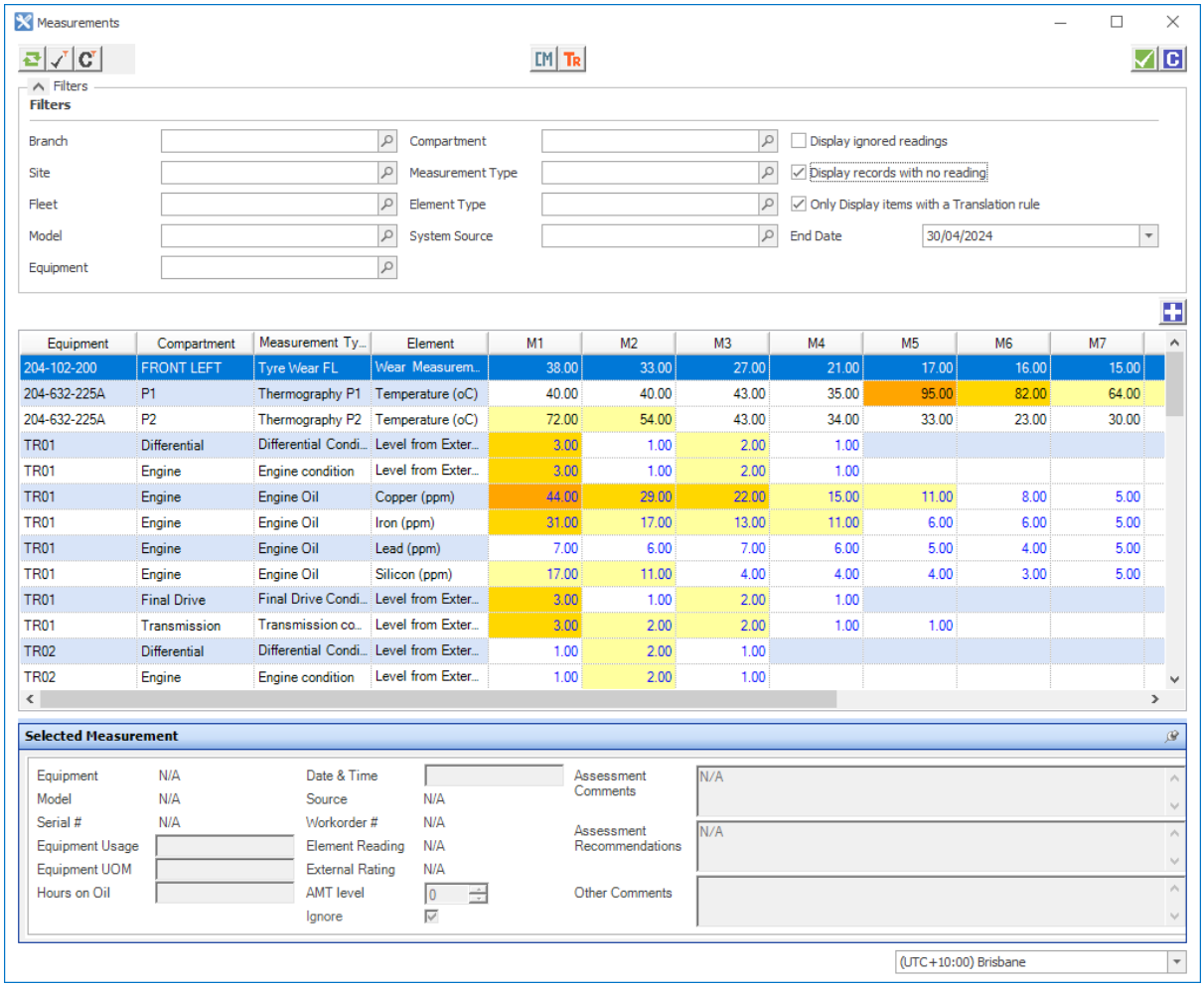
As part of the Work Finalisation process, measurements are typically updated in the work order. This section shows you how to manually update measurements in the Record Actual Measurement Readings table.

MEASUREMENT SETUP

Refer to the **RPM Training – Condition Based Maintenance** manual for information on measurement setup.

Procedure – Record Actual Measurements

1. Navigate **to Daily Operations** > **Usages & Measurements**, double-click **Record Actual Measurement Readings**.
2. Filter for the relevant assets, set **Display Records with No Readings** to **Yes** and click **Add**.





*You can paste values from a spreadsheet program into AMT, rather than entering the values one by one. Right-click on the Grid, select **Copy the Table**, paste the rows into the spreadsheet program, complete the empty cells, and then paste them back into AMT.*

After you've entered the reading, click **Refresh** to see how the measurement was classified.

Colour	Description
	OK
	Low Level Alert
	Medium Alert
	High Alert

Procedure – Analyse Actual Measurements

As a consequence of having measurements in AMT, you can trend this data:

- Consumption – Per UOM.
- Consumption – Per Month.

You can also report on the latest readings:

- Fleet Statistics.
- Equipment Statistics.

Exercise – Record Actual Usages

1. Record and interpret five actual measurement readings for an asset.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

45. Production Statistics

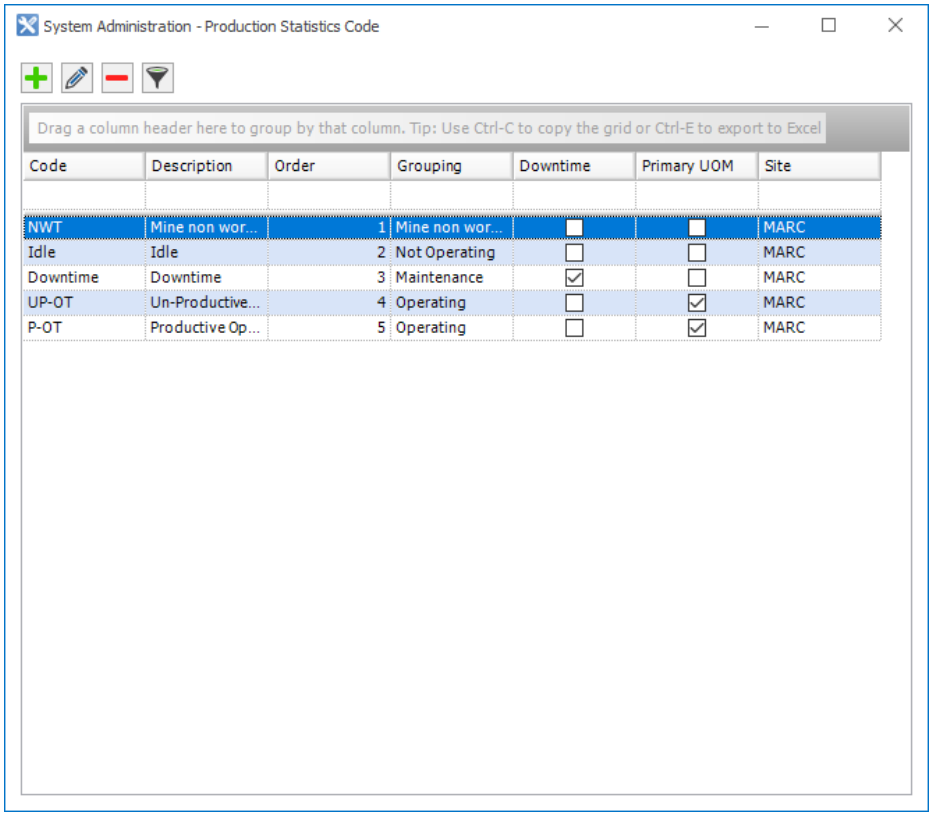
The **Production Statistics** function in AMT helps you capture productive and non-productive statistics for operational assets.

Actual downtime for an asset is sourced from the **Equipment Status** or the **Shiftlog**. Production statistics are used to capture the productivity detail for a machine so that you can understand and report on the utilisation of assets.

- Actual productive/non-productive statistics are entered into AMT by the admin clerk.
- A production downtime code is added to the production statistics.
- The production statistics utilisation rate is compared to AMT's utilisation, which is based on the primary UOM (such as engine hrs).
- Reports are generated which enable a better understanding of asset utilisation. This helps management to understand the problem areas, and to make corrective steps.

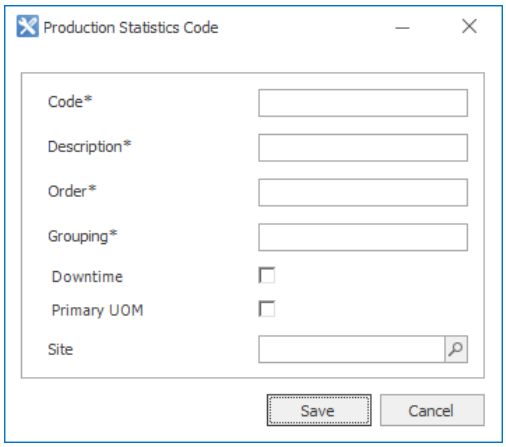
Procedure – Production Statistics Setup

1. Navigate to **System Administration** > **System Tables** > **Usages & Measurements**, double-click **Production Statistics Codes**.
2. Create **Production Statistics Codes** by clicking **+**. At least one record must be flagged as **Downtime**.



3. Complete all mandatory fields, **Code**, **Description** (for example, Weather Delay, Maintenance), **Order** and **Grouping (**for example, **Idle**, **Productive Utilisation**, **Non-Production Utilisation**).

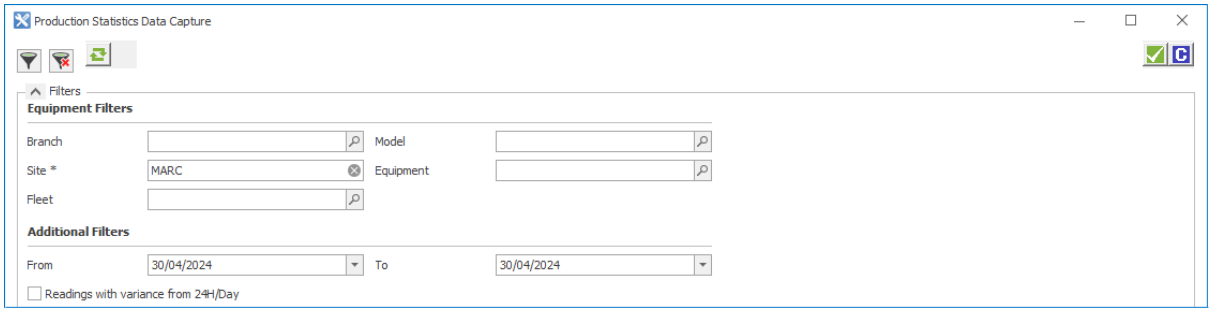
4. Other fields include:
    a. **Downtime**: Select this check box if the data for this code will be derived from actual downtime entered into AMT.
    b. **Primary UOM**: Select this text box if the code is based on AMT's primary UOM (for example, engine hours).
    c. **Site**: Select a site if the code is specific to one site.



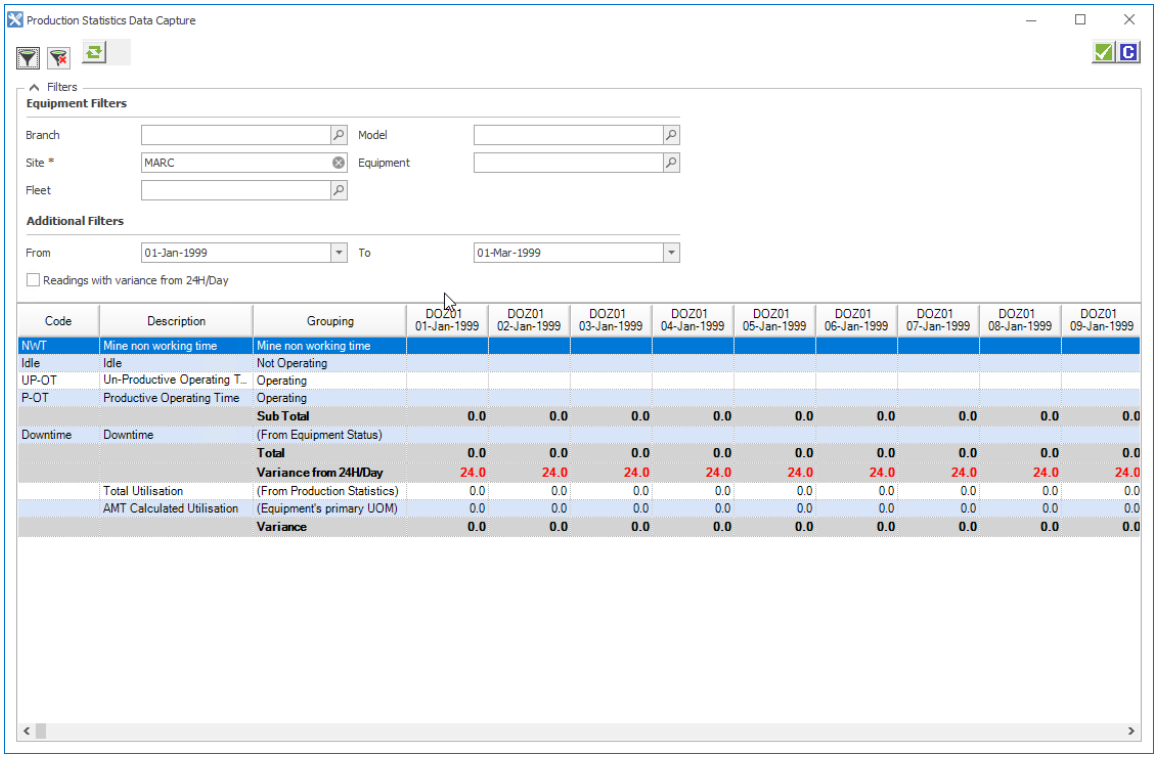
5. Click **Save**.

Procedure – Record Production Statistics

1. Navigate to **Daily Operations** > **Usages & Measurements**, double-click **Production Statistics Data Capture**.
2. Filter for the relevant **Site** and **Assets**.



3. Click on a grouping to add a new record.



Code	Description	Grouping	DOZ01 01-Jan-1999	DOZ01 02-Jan-1999	DOZ01 03-Jan-1999	DOZ01 04-Jan-1999	DOZ01 05-Jan-1999	DOZ01 06-Jan-1999	DOZ01 07-Jan-1999	DOZ01 08-Jan-1999	DOZ01 09-Jan-1999
NWT	Mine non working time	Mine non working time									
Idle	Idle	Not Operating									
UP-OT	Un-Productive Operating T...	Operating									
P-OT	Productive Operating Time	Operating									
		Sub Total	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Downtime	Downtime	(From Equipment Status)									
		Total	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
		Variance from 24H/Day	**24.0**	**24.0**	**24.0**	**24.0**	**24.0**	**24.0**	**24.0**	**24.0**	**24.0**
	Total Utilisation	(From Production Statistics)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	AMT Calculated Utilisation	(Equipment's primary UOM)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
		Variance	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**

The Maintenance Downtime is showing null values, which indicates there is no event in AMT and no downtime has been captured for the completed work. This may occur on site if the production statistics data is captured on one form, and the downtime is captured on a **Shiftlog** or **Breakdown** form. It is also possible the asset wasn't down for any scheduled or unscheduled maintenance during that period.

- The **Variance** shows the difference between the total productive/non-productive utilisation and a 24 hr period.
- **Total Utilisation** (from production statistics) = productive or non-productive utilisation which is based on the grouping.
- **AMT Utilisation** is based on the asset's primary UOM, such as engine hrs.

After you've updated the downtime record in AMT, this figure will be reflected in the production statistics. If a mistake is made, correct the entry by editing the hours with a 'zero value' and then saving.

Data in this table can be copied to Excel, edited and pasted back into AMT. Right-click the grid to copy the data.

Procedure – Analyse Production Statistics

There are two reports in AMT that provide accurate reporting for productive and non-productive work time, which helps understand asset performance:

- Production Statistics Analysis.
- Production Statistics Chart.

Exercise – Record Production Statistics

1. Record and interpret production statistics for 5 assets.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

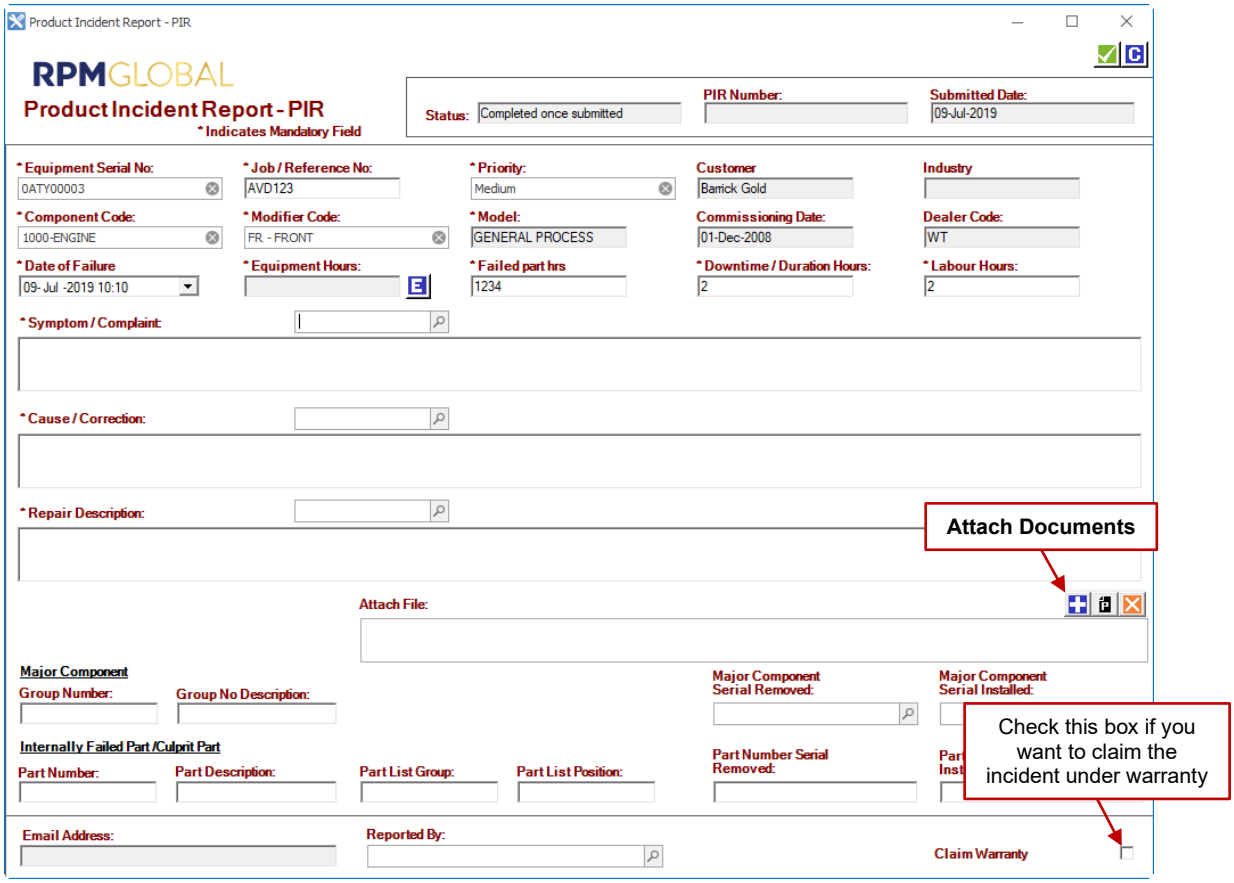
Training Workshop (Product Incident Capture & Warranty Claims)

AMT has a form that provides a structured process for registering and processing Product Incident Reports (PIR) and for registering and processing warranty claims.

46. Registering Product Incident Reports

Procedure – Registering Product Incident Reports

1. Navigate to **Daily Operations** and double-click **Product Incident Report Capture**.
2. Complete all mandatory fields (marked with an asterisk *).
3. **Symptom**, **Cause** and **Repair** are details on why the component failed. Failed Part information is at the bottom of the screen.
4. Attach electronic files (such as failure reports, photographs) to the PIR by clicking **Add**. AMT will copy the documents from your computer and save them to the AMT server.
5. If you want to claim the incident under warranty, click **Claim Warranty** at the bottom right of the screen. The **Claim Warranty** flag does not mean approval. Rather, it indicates to head office that a warranty claim has been registered in AMT that needs to be reviewed.

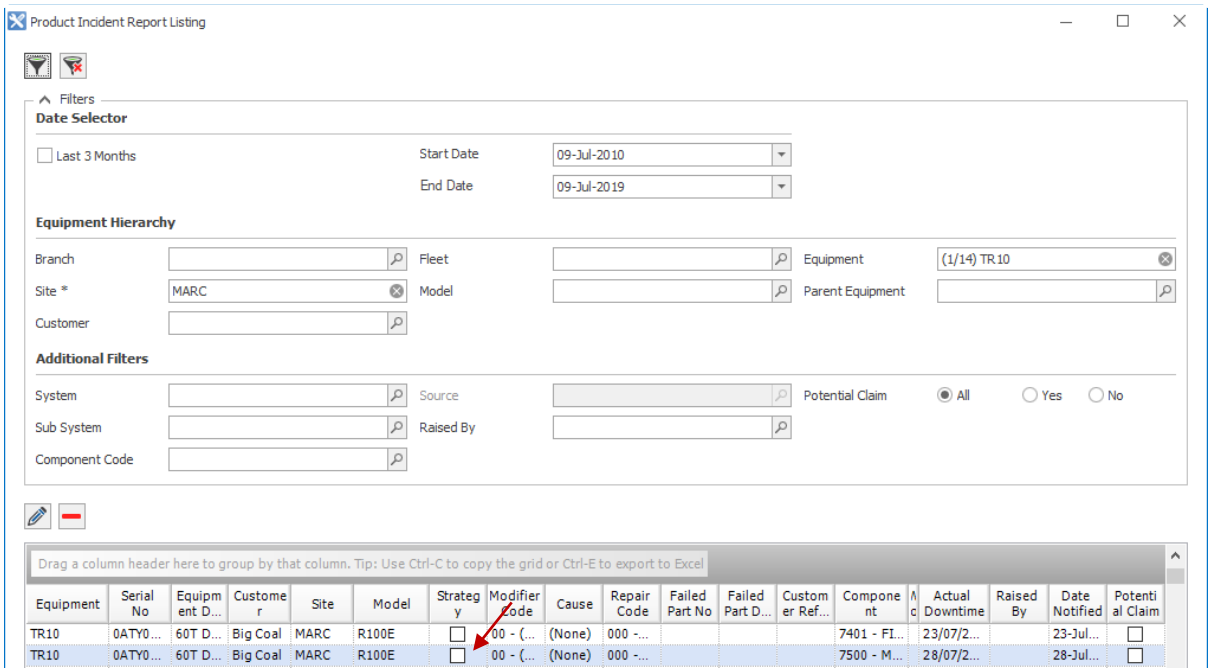


6. PIRs saved in AMT are recorded as work orders.
7. The work orders are automatically assigned a **Source** from the AMT Source system table, where the **Source** is flagged as a **Customer Reports Issue**.
8. The asset must be registered in AMT for you to create a PIR for it.

47. Reviewing Product Incident Reports

Procedure – Reviewing Product Incident Reports

1. Navigate to **Daily Operations**. Double-click **Product Incident Report Listing**.
2. Filter for a **Site**.
3. Select a **Date Range**.
4. Click **Apply Filters**.
5. In the grid, double-click a record to open the **PIR Capture** screen.



*The grid can be copied to a spreadsheet program by right-clicking and selecting **Copy**.*

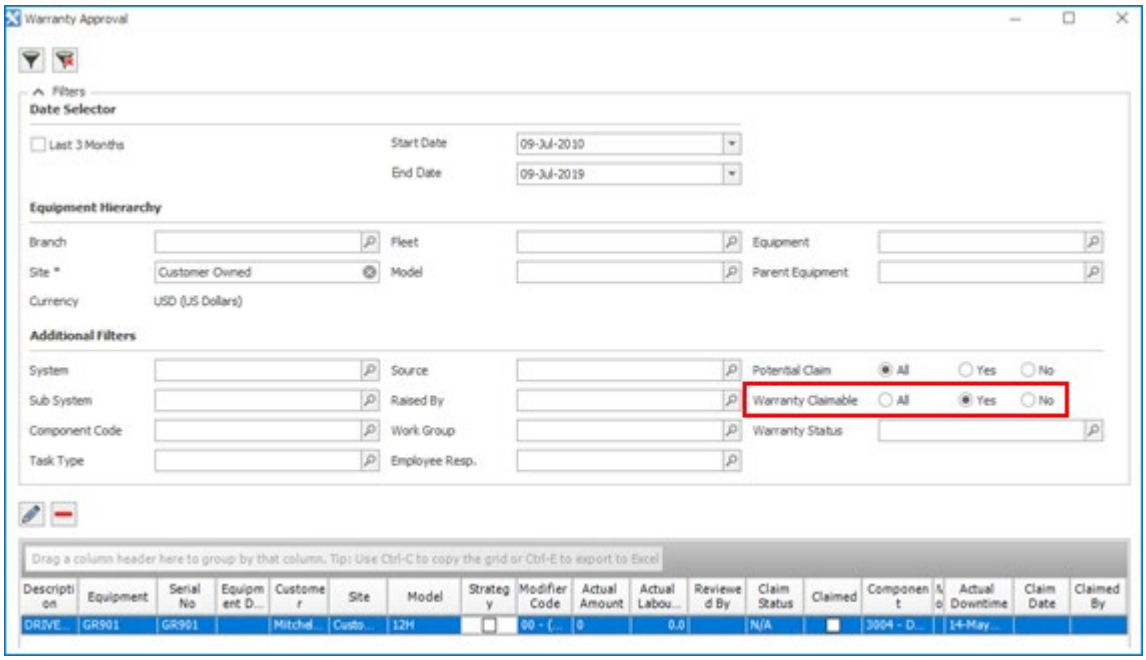
Exercise – Create a PIR

Create and review a product incident report.

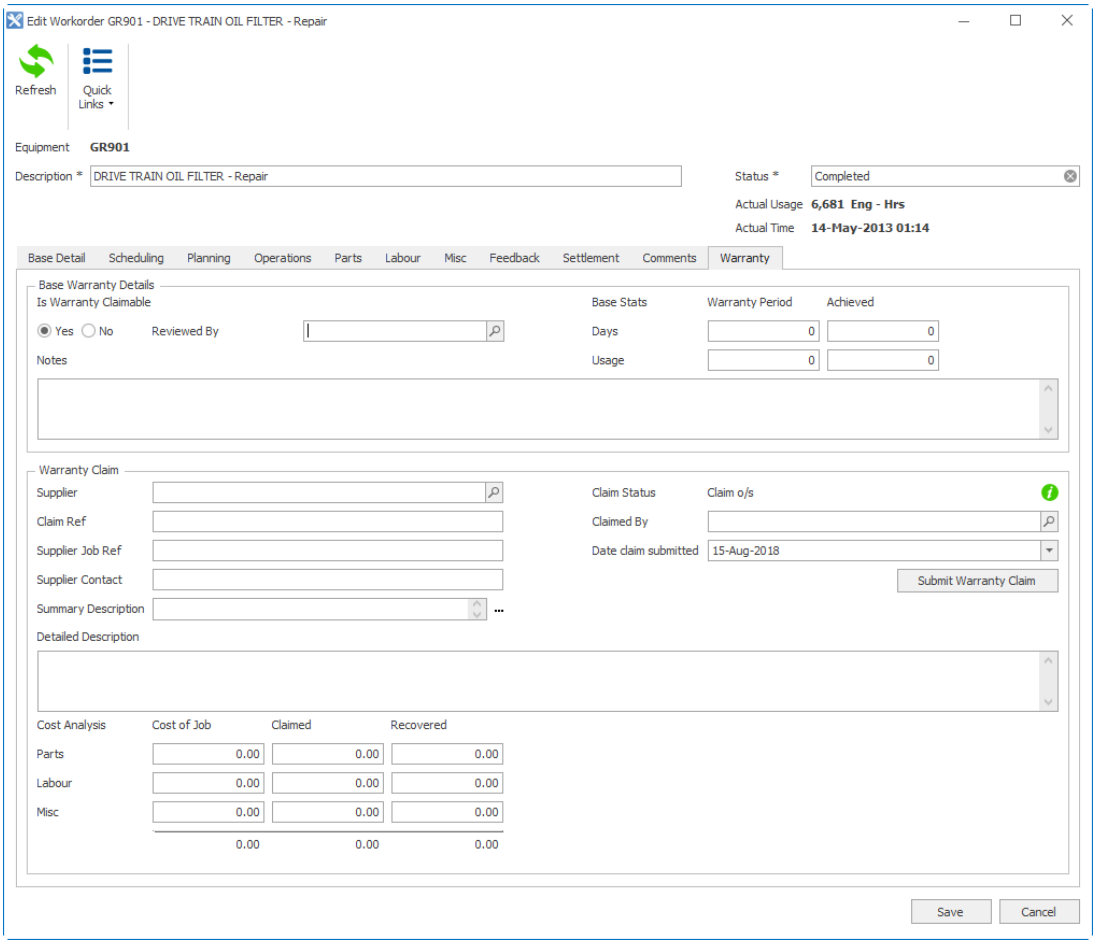
48. Assess Warranty Claims

Procedure – Assess Warranty Claims

1. Navigate to **Daily Operations** > **Warranty**, double-click **Warranty Approval**.
2. Select a **Site** and any other filters.
3. Select a **Date Range**.
4. Set **Warranty Claimable** to **Yes**.
5. Click **Apply Filter**.
6. In the grid, double-click on a row to show the applicable work order.



7. Search for work orders that have **Warranty Claimable** = **Yes**. Review the details of the work order by highlighting the work order and clicking **Edit**. The **Workorder Editor** screen opens, showing the **Warranty** tab.



8. On the **Warranty** tab, enter additional details about the warranty claim. A work order can have the following claim statuses:

   a. **N/A**: if the work order is marked as not claimable.

   b. **Claim o/s**: if the work order is marked claimable, but no claim has been submitted.

   c. **Claimed**: if either an amount is entered in the **Claimed** field (in the **Cost Analysis** section) or a claim has been submitted through the interface (that is, by clicking **Submit Warranty Claim**).

   d. **Closed**: if an amount is entered in the **Recovered** field (in the **Cost Analysis** section).



Flagging a work order as claimable is not limited to the PIR process. A standard AMT user can flag a normal, everyday work order as claimable.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

49. Submit Warranty Claims

If the warranty interface is in place, you can submit a warranty claim to a warranty system by clicking **Submit Warranty Claim** and selecting the warranty system to submit the claim to. AMT will then:

- record the date, time and username related to the request.
- send a request to the selected warranty system to process the warranty claim. This claim is processed in real-time order to provide live feedback to the user.

If the warranty claim is denied, the warranty system will send an error message back to AMT, and the claim status will change to **N/A**. The error message will appear in a message box. Click **Details** to read why the claim was denied. The text contained in the message box is provided to AMT by the warranty system.

If the warranty claim is approved, the claim status changes to **Claimed**.

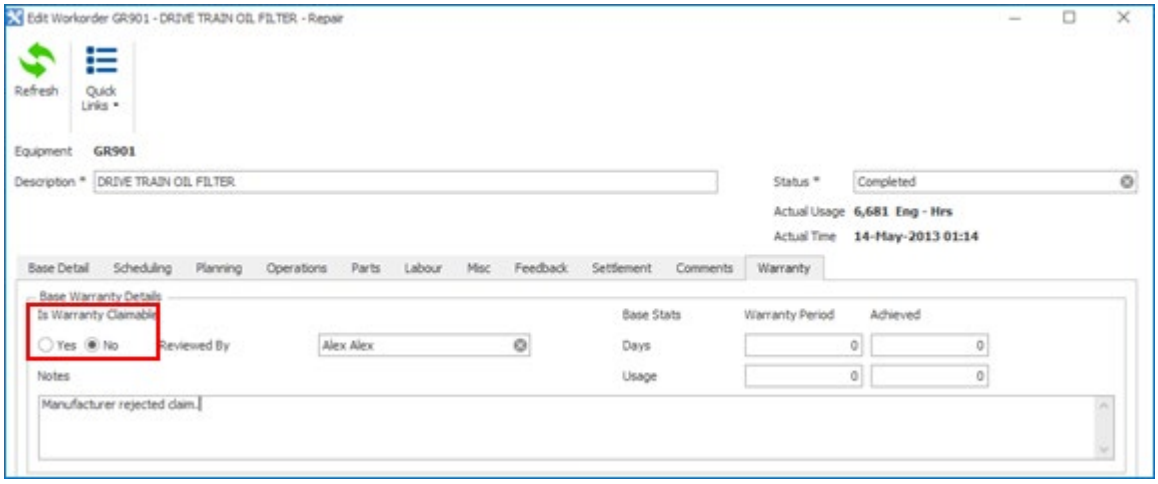


This only works if the warranty interface is in place.

50. Not Claimable Warranty Claims

Procedure

1. To manually flag the work order as not claimable, set **Warranty Claimable** to **No** (notice the **Claim Status** changes to **N/A**).



2. Click **Save**.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes